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RECOMMENDATION REPORT



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Message to Executive

Dear Executives of mDetect,

Please find enclosed our report, which has been prepared for providing you with advice in relation to EduMOD's international expansion efforts. Based on our analysis of the United States market, we have identified the challenges concerning the bottleneck in the distribution that you are currently experiencing, paired with the difficulties in acquiring clients due to your current pricing strategy as well as the lack of localisation and social media presence.

Our primary recommendation is to target high school districts in California by establishing a subsidiary in San Jose and facilitating communications through handwritten letters, email campaigns, social media campaigns, and subscriptions/lease models for the EduMOD. This recommendation has been made to position the EduMOD effectively within the K-12 education sector while addressing the bottleneck in the distribution channel and limited social media presence.

To substantiate our recommendation, our propositions are as follows:

- Develop dedicated educational applications with interactive elements, gamified progression and varied question types.
- Develop a three-tiered pricing model: one-time purchase, five-year lease or monthly lease for the EduMOD.
- Establishment of a subsidiary in San Jose.
- Marketing strategies involving trigger-based automation email campaigns and handwritten letters, social media marketing by utilising channels such as Youtube, LinkedIn, Instagram; and grassroots advocacy campaign via interactions with Parent-Teacher associations (**PTA**).

Sincerely,

McSwinney Consulting

Executive Summary

This document aims to address and provide recommendations for mDetect Pty Ltd in the international expansion of its flagship product, the EduMOD, which uses muon tomography technology for educational purposes. The scope of this document is relegated to international entry into the United States market due to its lucrative pool of educational funding and close alignment with the EduMOD's value proposition.

We have identified that mDetect experiences a three-fold problem in entering the United States market:

- Sole reliance on cold emails and cold approach methods which dampens mDetect's brand recognition because it does not utilize all possible channels, affecting the CEO/Founders of mDetect by making US a less appealing market.
- Expensive pricing (i.e 3,392.5 AUD or 2100 USD) relative to other competitors in the market such as CosmicWatch or United Kingdom Radio Astronomy Association (**UKRAA**), affecting the channel partners, potential clientele and shareholders as purchase contracts would be more likely to fall through.
- Limited distribution channels due to the dwindling size of the school retail wholesaling industry (Diment, 2025), affecting mDetect's likelihood of finding a channel partner and the partner's likelihood of accepting distributions of the EduMOD.

Our analysis of the external market in the United States reveals that there are significant opportunities in targeting school districts in California and Texas due to their large budgets and STEM funding from the US Government, and adherence to the California Scientific Framework, which emphasises practical application of science.

However, the recent political controversies involving tariffs from the Trump administration and the remaining favouritism towards traditional textbooks may present a challenge to EduMOD's successful entry into the United States.

As such, the target segment is identified as school districts in California (primary segment) and Texas (secondary segment). This segment is challenged with low-impact educational technology options and seek Next Generation Science Standards (**NGSS**) aligned options that boost student engagement.

Given these analyses, our recommendations are as follows:

- Develop dedicated educational applications with interactive elements, gamified progression and varied question types with adjusted curriculum for geology for the US.
- Develop a three-tiered pricing model: direct sale, one year lease, 2-year lease, or lease-to-own strategy.
- Establishment of a subsidiary in San Jose following a Limited Liability Company (**LLC**) with generic shipping from the head company in Australia to the subsidiary.
- Marketing strategies involving trigger-based automation email campaigns and handwritten letters, social media marketing by utilising channels such as YouTube, LinkedIn, Instagram, and grassroots advocacy campaigns via interactions with Parent-Teacher associations.

With this strategy, we estimate that the subsidiary would require \$339,202 for initial set-up and three years until the subsidiary would break even in Year 3 and achieve a return on investment of 52% by year 5, given flat sales growth of 15%. The estimation is due to optimistic projections on future outreach to cities like San Francisco because of widespread STEM funding in California.

Overall, this recommendation is suitable for the EduMOD's value proposition by appropriately addressing the target market's concerns by providing a complete educational package that emphasises practical scientific application. The development of an application for learning materials further addresses the target market's concerns by meeting rising demand for engaging, hands-on learning experiences.

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I. Introduction

The purpose of this hereby enclosed report is to address the broader challenges that mDetect would likely experience in its efforts to expand the EduMOD into the education sector in the United States.

We aim to identify potential challenges through utilising the 5H IWs, and the Ishikawa Fishbone diagram, and uncover the root cause that may affect your internationalisation efforts in the United States.

This report will also analyse the external business environment, identifying customer segments, evaluating competitors using tools such as PESTLE, Porter's Six Forces, the 3Cs (Customer, Competitor, Company) diagram and subsequently a SWOT analysis.

This will evaluate internal strengths and weaknesses of your firm's internal functions, as well as identify opportunities for the strategy, which will subsequently be used to analyse the broader challenges that mDetect may face in entering the U.S Market. The relevant data sources utilised will be market data for the education sectors in the United States and relevant macroeconomic and microeconomic data in identifying these challenges

Recommendations will be provided based on these analyses using market research and accounting techniques such as income projection and balance sheet projection. The recommendation will cover several key areas, such as pricing strategy, distribution strategy; marketing and communication strategy. The recommendations will be evaluated using the Suitability, Acceptability, and Feasibility (**SAF**) framework evaluate the appropriateness of our recommendations.

2. Problem Overview

The core issue lies in mDetect's current distribution channel to the United States, paired with an inefficient pricing strategy and limited social media presence, as mDetect solely relies on a cold approach using emails and scientific exhibits to attract clients, and lightly utilises the social media channels to boost its brand recognition.

The following Ishikawa Fishbone Diagram (Kumah et al., 2024) illustrates the root causes:

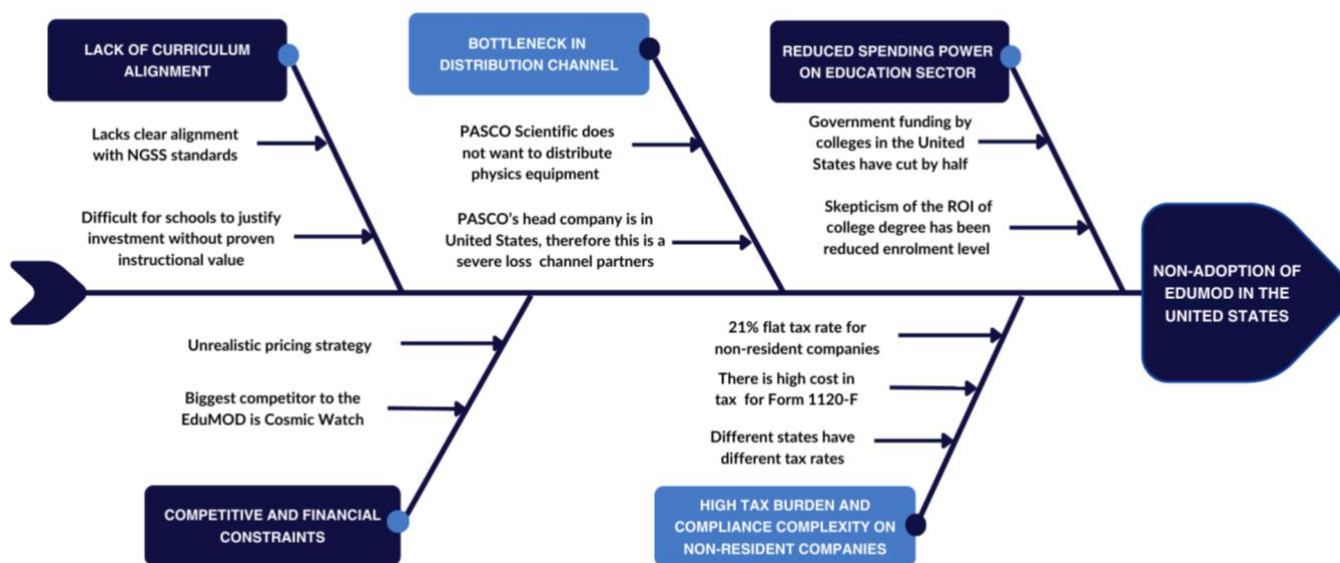


Figure 1. Ishikawa Diagram regarding root causes

The issue, therefore, affects several key stakeholders, such as the CEO/Founders of mDetect, as well as the shareholders of mDetect, high school boards and universities, as well as the channel partners in the USA.

The CEO/Founders of mDetect and the shareholders of mDetect would be more likely to face a higher risk of not being able to break into the USA market without a discernable online presence. A study by Dwivedi et al. (2021) found that over 88% of businesses use Twitter for marketing purpose. This means that not having an online social media presence damages the CEO as they would lose investor confidence and end up misallocating resources to the United States market.

According to Suryati and Lusia (2022), distribution channels have a positive correlation with sales volume, meaning that a more effective distribution channel is correlated with a higher sales volume. As such, experiencing bottlenecks in the distribution channel materially affects mDetect themselves as the channel partner's decisions mean that the EduMOD would not be able to break into the U.S market.

EduMOD's high pricing strategy has had significant consequences for three key stakeholder groups in its penetration of the US education market, including school boards and administrators, distribution partners, and mDetect itself. For distribution partners, high prices reduce competitiveness and make it difficult to market products, especially when compared to domestic competitors that are priced up to 21 times lower (Rajagopal, 2016).

EduMOD's lack of clear alignment with popular K-12 standards like the Next Generation Science Standards (NGSS) also undermines the perceived value of the product (NGSS, n.d; Wyssession, 2014). However, without a product that can demonstrate compatibility with NGSS, it will be difficult for school district boards to justify a high investment if the instructional benefits are not clear or immediate.

3. External Environmental and Market Analysis

To comprehensively grasp the scope of the target market, this report will deploy the 3Cs (Customer, Competitor, Company); Porter's Six Forces; PESTLE; which will be distilled into a Strengths, Weaknesses, Opportunities, Threats (**SWOT**) matrix to inform our recommendations.

3.1. 3Cs Model

3.1.1. Customers – K-12 school districts

- **Prospective customers:** K-12 school district boards (in populated states like California, and dense high high-quality school districts in Cali like Palo Alto Unified School District, Mountain View-Los Altos Union High School District, etc.).
- In California, a total of 6.86 million students are enrolled in K-12 schools (Bouchrika, 2021a). This is the highest number of K-12 students in the U.S.
- In Texas, a total of 5.84 million students are enrolled in K-12 schools (Bouchrika, 2021).
- In California, most school districts adhere to the California Science Framework (**CSAF**) (California Department of Education, 2016).

3.1.2. Competitor

EduMOD utilises muon detection technology to facilitate students' exploration of complex scientific ideas such as cosmic rays, special relativity, and particle physics. Despite its distinctive muon detection capabilities and learning materials, EduMOD encounters competition from entities such as CosmicWatch, the UK Radio Astronomy Association (UKRAA), and PhET, which provide comparable teaching instruments (Refer to [Appendix A](#) for competitor strengths).

	Cosmic Watch	UK Radio Astronomy Association (UKRAA)	PHET
Value Proposition	Simple, physics-motivated machine and electronic-shop project for university and schools. Can be used in conjunction with provided software to make physics measurement. (CosmicWatch, 2017)	Readily assembled and tested muon detector. Enclosure sold separately. (UK Radio Astronomy Association (UKRAA), 2025)	Online science simulations platform made by the University of Colorado Boulder specifically for the classroom The platform's purpose is to create an initiative to improve quality of global math and science (University of Colorado Boulder, 2019)
Pricing	\$100 USD	278 USD for each model with enclosure. (UK Radio Astronomy Association (UKRAA), 2025)	Free
Packaging and distribution method	N/A, CosmicWatch only provides a github guide along with andruing codes. (CosmisWatch, 2017).	Orders are restricted to a maximum of 4 items, any larger quantity requires contact for a quote. International clients are responsible for payment of duties imports or VAT. (UK Radio Astronomy Association (UKRAA), 2025)	N/A, the platform is online
Client Acquisition Method	N/A, CosmicWatch does not have a business model and relies on articles such as Chu (2017) to spread its brand recognition.	Limited social media presence as UKRAA is a nonprofit entity. The entity appears to acquire clients via the store website	N/A, the platform is online
Comparision to EduMOD	There is no discern able distribution network in sight. There is no simple software to review the results like the EduMOD This is unfit for the CSAF framework, given that the assembly manual is very complex.	Limited distribution channel and cannot scale, unlike the EduMOD. Cheaper price point, but no software access to view r Limited distribution channel and cannot scale, unlike the EduMOD. Might pose a threat to EduMOD, however their operation scale is too small in relation to EduMOD.	Schools would be keen to adopt the platform given it is free. However, still unfit with the CSAF framework since the framework also requires practical experiments. Therefore, this might pose a substitution threat to EduMOD.

Table 1. Competitor Comparison and Analysis

Based on the identified competitors, a perceptual map can be drawn using horizontal axis indicating the price point of a product, and a vertical axis that quantifies the degree of quality and attributes integrated into the product (See [Appendix B](#) for brand positioning explanation).

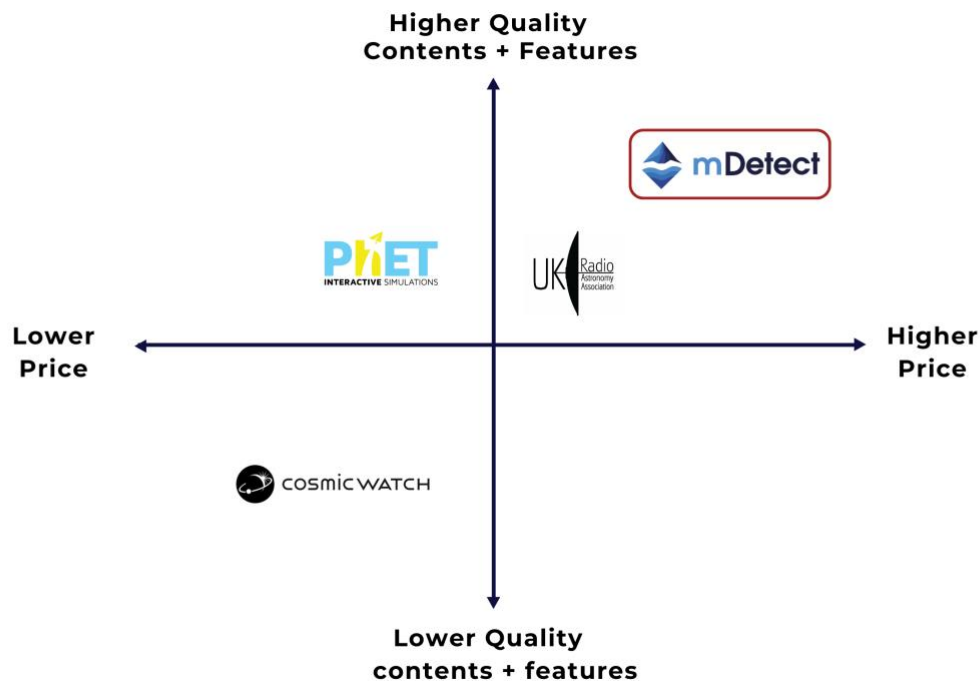


Figure 2. Perceptual map

This perceptual map illustrates EduMOD's branding as a premium brand, providing superior content and advanced features at a higher pricing point. This makes it more attractive to consumers who value practical learning and are willing to spend in a more advanced product. Conversely, brands such as CosmicWatch and PhET cater to the budget-conscious demographic by providing more economical alternatives with moderate content and features. UKRAA's product, situated in the medium-high quality segment at a premium price, is designed for universities and research facilities, targeting clientele that prioritise scientific accuracy and are willing to invest in a specialised physics instrument.

3.1.3. Company (mDetect)

- **Vision of EduMOD:** to teach students about concepts ranging from special relativity to statistics and exploration of the high-energy universe and nuclear physics in a hands-on manner.

- **Current internationalisation strategy:** Via channel partners who distribute the EduMOD into target markets. Currently distributed to the USA, UK, Indonesia, Vietnam, Japan.
- **Brand recognition:** Has been known locally through channel partners, and mDetect relies on word-of-mouth as the main method for brand recognition. Limited social media presence.
- **Client acquisition model:** Mainly through cold emails, scientific expos and communication with physics teachers who pitch the products to school boards. Social media marketing is largely underutilised.
- **EduMOD product offering:** The EduMOD package comprises a single-pixel camera-based muon detector (EduMOD version) designed for classroom application, along with cables connecting the device to an external laptop or desktop. A web-based graphical user interface (GUI) facilitates remote control of EduMOD via a laptop or desktop, ensuring user-friendliness.

3.1.4. 3Cs Analysis Model Mapping

Based on the above information, we have constructed a 3Cs map that evaluates EduMOD's position in the educational technology market as follows:

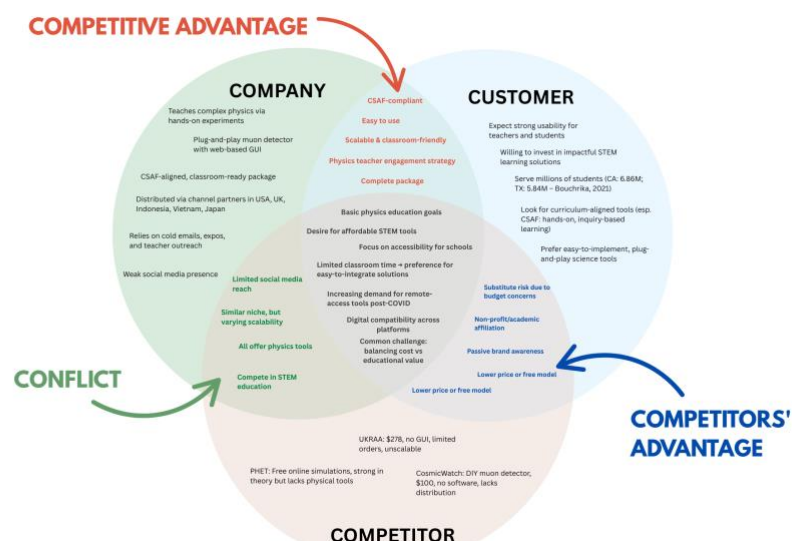


Figure 3. 3Cs Analysis Mapping

EduMOD presents a significant competitive edge by providing practical physics instruction via its plug-and-play muon detector, which is CSA-compliant and simple to integrate. Its emphasis on physics education and teacher involvement method establishes it as a holistic, scalable solution within the STEM education sector. EduMOD encounters considerable competitive pressure from businesses like PHET and CosmicWatch, which provide lower-cost or complimentary devices and possess greater brand recognition and accessibility.

EduMOD's dependence on cold outreach and inadequate social media presence results in a disconnect with client expectations for robust usability and accessible, easily implementable technologies. Although EduMOD aims for the high-end market with its premium features, it faces challenges in fulfilling the demand for cost-effective STEM solutions, which may restrict its market share.

3.2. Porter's Six Forces

Force	Force assessment	Details
Power of Buyers	High	Districts purchase through consortia like Ed Tech JPA, giving buyers strong leverage to demand curriculum fit and cost-effectiveness (Education Technology Joint Powers Authority, 2025). EduMOD's niche focus on muon detection lacks clear alignment with NGSS, making it harder to justify in procurement decisions (California Department of Education, 2016).
		→ Power of buyers in the California market is high due to the state's centralised and standardised buying process and the existence of more relevant products. Without tighter curriculum alignment and local proof of impact, EduMOD risks early rejection in bulk procurement processes.

Power of Suppliers	Low	Suppliers identified: Every muon detector is made by plastic scintillator viewed by a photomultiplier tube (University of Adelaide, 2023). Number of suppliers in the market: The only publicly available information is that the market size in 2024 is 192 million USD, which is expected to grow to 281 million in 2023 (Imarc, 2025). The supplier market appears to be quite sizeable. Some of the major players in scintillator market include Canberra Industries Inc. (U.S.), Inc. (U.S.), Mirion Technologies Inc. (U.S.), Radiation Monitoring Devices (U.S.) (MarketsandMarkets , 2020). → Power of suppliers are low due to the rapid development and large size of the scintillator markets in the US. EduMOD may have the freedom to explore the markets for suppliers that offer the most cost-efficient deals.
Risk of Substitutes	Moderate to high	Alternative Science subjects: Schools may allocate limited resources to STEM subjects more prominently featured in state standards, such as environmental science, biology, or renewable energy. These areas align more closely with standardized assessments and performance metrics outlined in California's NGSS-aligned 2016 Science Framework, where particle physics is not explicitly prioritised. (California Department of Education, 2016) Preference for traditional textbook over hands-on tech: A 2024–25 budget report shows that Palo Alto allocated \$8.6 million for instructional materials, suggesting a continued prioritisation of traditional learning tools (Palo Alto - Unified School District, 2024). → EduMOD may face stiff competition from traditional resources, which are not only cheaper but also more familiar to staff. It would require tremendous efforts from EduMOD to convince schools to adopt their technology in the starting phases.

Risk of New Entrants	Moderate to High	Massive STEM Education Investment: California boasts the largest education budget in the U.S., with over \$21.7 billion allocated, including dedicated funds for NGSS implementation and STEM innovation grants (Legislative Analyst's Office, 2023). California follows a market-driven regulatory model. This allows new and adjacent entrants to launch, test, and scale quickly without needing extensive approvals — creating a dynamic, innovation-driven landscape. As highlighted by McKinsey, companies that iterate rapidly and capture early partnerships often outperform slower-moving competitors (Bradford, 2023). ➔ California has one of the most exciting start-up ecosystems, which opens the doors for new competitors. While this open the door for many funding opportunities, it also requires EduMOD to have a sustainable strategy so that it would not be outgrown by faster and leaner competitors.
Competitive Rivalry	Medium	Direct competitors and price pressure • CosmicWatch (MIT): An undergraduate-level project from MIT, showing users how to build their own muon detectors priced at \$100. However, the project only provides guides and an Arduino code, not a direct product. (Chu, 2017) • DIY Muon Detector Kit (Elecrow/Hackaday): Priced around \$100, this kit allows users to assemble their muon detector. It appeals to individuals or small organizations seeking affordable technology. (Posch, 2025). ➔ The competitive muon detectors are just self-assembled kits and are still in the development stages, with little to no direct relevancy in the education sector. EduMOD needs to exploit this “golden phrase”—when most of the competitors are lagging far behind to establish its reputation.

Complementary Product	Low	EduMOD Package: • Single Pixel Camera-Based Muon Detector (EduMOD version) - built with advanced SiPM technology for safe and accurate detection of high-energy particles • Web-Based GUI - allows students and teachers to interact with the detector through any standard browser on an externally sourced laptop or desktop (no downloads required). • Browser Compatibility is high, but does not introduce risk in terms of needing other complementary products. Learning and Teaching Resources: • Curriculum-aligned educational resources, developed specifically for Year 11, Year 12, and undergraduate physics
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Table 2. Porter's Six Forces Analysis

3.3. PESTLE Analysis

P Political	• Tariff Threats Ahead: Upcoming reciprocal tariffs set out by the Trump administration threaten to set tariffs of 10% on products coming from Australia (Koziol, 2025; Department of Foreign Affairs and Trade, 2025). • Massive Education Grants: The Department of Education has disbursed 179.1 billion USD to state and education agencies according to 2024 fiscal year reports, with grant expenses totaling 150 billion (U.S. Department of Education, 2024). • Strong US–Australia Ties: Australia and the United States are reliable trade and investment allies, engaging in US\$77 billion of bilateral trade and maintaining an investment relationship of US\$1.6 trillion, making the U.S. Australia's foremost economic partner (Nicholas Kittel, 2017).
	Application The prospective reciprocal tariffs on Australian goods present a risk to EduMOD, potentially increasing expenses. Nonetheless, robust commercial relations between Australia and the U.S. present an opportunity to alleviate this through bilateral agreements (Kittel, 2017). Simultaneously, the Department of Education's investment of US\$179.1 billion for education offers EduMOD a chance to obtain funding and enhance its market position (U.S Department of Education, 2024).

E

Economic

- **Shrinking Disposable Income:** Previously, escalating disposable incomes enabled households to accommodate rising tuition expenses. In recent years, this trend has diminished as inflation has diminished real income, escalated the cost of necessary products and made prospective students more attentive to prices in their higher education choices (Seiler, 2025).
- **Eroding School Budgets:** School budgets face potential erosion: While the annual inflation rate eased to 2.8% in February 2025, core inflation was 3.1%, and forecasts suggest a possible rise later in the year, which could strain school purchasing power (Trading Economics, 2025; Sherter, 2025; USA Facts, 2024).
- **Slower Growth Risks Funding:** Economic slowdown could tighten public spending. Anticipated slower US economic growth by the end of 2025 might lead to reduced state and local government budgets, impacting education funding (Dwyer, 2025).
- **Record K–12 Funding:** However, recent increases in state expenditure and singular federal pandemic allocations have resulted in unparalleled funding levels for K–12 education (Murphy & Paluch, 2018).



Application

Inflation and fiscal constraints render schools increasingly price-sensitive, jeopardising EduMOD (Seiler, 2025; Trading Economics, 2025). The economic recession and possible reductions in state and municipal funding intensify these issues, posing an additional threat (Dwyer, 2025). Recent pandemic relief funding has resulted in unparalleled K–12 expenditure, creating a transient opportunity for EduMOD to assimilate into educational institutions (Murphy & Paluch, 2018). These considerations underscore the equilibrium between fiscal constraints and government-induced opportunity.

S

Social

- **STEM as National Priority:** Americans place a high value on being a global leader in scientific achievement and observe favorable outcomes from government investments in scientific research (Funk et al., 2020).
- **Parental Support for STEM:** Young children and their parents exhibited predominantly favorable attitudes towards STEM education. Many parents recognised the economic importance of STEM education for their families, believing it could lead to future financial wealth for their children (Wan et al., 2020).
- **Academic Performance Gap:** The academic performance of students in the U.S. in science and mathematics has lagged behind that of students from other countries for over a decade (Gillespie, 2021).
- **STEM Media Influence:** Digital media are prevalent in the lives of youth and offer opportunities for students to acquire knowledge in STEM. Numerous hypotheses contend that the STEM media landscape may influence youths' perceptions of prospective STEM careers (Chen et al., 2023).
- **Rising Educational Demand:** Growing US population signals increasing educational demand: The US population is projected to reach around 347–350 million in 2025, potentially creating a larger market for educational resources (Congressional Budget Office, 2025).



Application

The rising focus on STEM education and its economic significance to parents creates an opportunity for EduMOD, as the need for interactive STEM tools is anticipated to rise alongside growing populations and changing educational requirements (Wan et al., 2020; Congressional Budget Office, 2025). The historical inadequacy in the scientific and mathematical performance of U.S. pupils (Gillespie, 2021) presents an opportunity for EduMOD to rectify this disparity through its interactive approach. The proliferation of digital media and students' growing inclination towards interactive, technology-based learning provide an opportunity for EduMOD to connect with this trend and support the United States' STEM leadership objectives (Chen et al., 2023; Funk et al., 2020).

T

Technological

- **Growth of Online Learning:** Currently, online learning platforms such as Blackboard, Brightspace, and Moodle, along with classroom management tools, are gaining popularity due to the rise of asynchronous learning in the US, particularly among K–12 students. As device usage increases and students become more accustomed to online learning, school districts are shifting towards more flexible schedules (Torchia, 2022).
- **Innovative STEM Trends:** Emerging technology trends in STEM education include AI-powered learning, Social and Emotional Learning (SEL), screen-free technologies, and a growing focus on sustainability and climate change education (Spector, 2024).
- **EdTech Standards Compliance:** The U.S. maintains strict standards for educational technology, with the ISTE Standards (International Society for Technology in Education) being among the most recognised. These standards guide how technology should be used to create effective and equitable learning experiences (Nelson, 2023).
- **Strict Data Protection Laws:** The US enforces strict regulations on data security and privacy in education, including:
 - COPPA – Requires notice and verifiable parental consent for data collected from children under 13 (Schifferle, 2020).
 - FERPA – Protects student education records and restricts disclosure without consent (Student Press Law Center, 2024).



Application

The growth of online learning offers an opportunity for EduMOD to expand its market, though increased competition presents a threat (Torchia, 2022). Emerging STEM trends like AI-powered learning are an opportunity for innovation, while screen-free technologies may pose a threat (Spector, 2024). ISTE Standards ensure product effectiveness but create a challenge in terms of compliance costs (Nelson, 2023). Strict data protection laws represent a threat in compliance costs but also offer an opportunity to build trust and enhance EduMOD's position.

L

Legal

- **FERPA Compliance Required:** Data Protection and Privacy: EduMOD must comply with FERPA (U.S. Department of Education, 2018).
- **Public Procurement Rules:** School Procurement Compliance: To sell to public school districts in California and Texas, EduMOD must follow public procurement rules, including vendor registration, fair bidding, and meeting ethical sourcing standards (Carreon & Fagen Friedman and Fulfrost LLP, 2023).
- **Labor Law Obligations:** Employment and Labor Regulations: If hiring staff locally, mDetect must comply with U.S. labor laws (e.g., FLSA), ensure safe working conditions, and meet background check requirements for employees working in school environments (Federal Register, 2024).



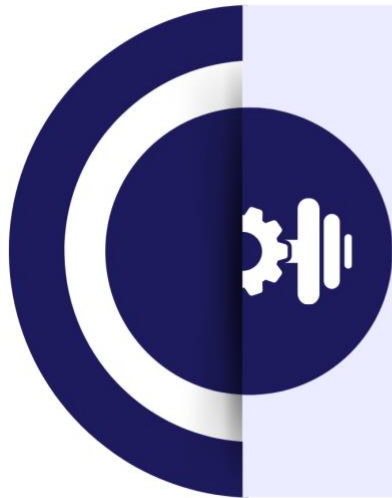
Application

FERPA compliance is an opportunity for EduMOD to establish trust with educational institutions through the secure management of student data, hence facilitating access to the U.S. education sector. Procurement restrictions in California and Texas present an additional possibility, facilitating access to substantial school districts and broadening EduMOD's market penetration. Adherence to U.S. labour rules is an opportunity, bolstering EduMOD's reputation as an ethical employer and drawing in proficient personnel. Nevertheless, compliance with California's accessibility standards (WCAG 2.1) poses a challenge, as it necessitates considerable resources for ongoing adherence, thereby escalating operational complexity and expenses.

E Environment	• Tailings Dam Stability Monitoring: Muon detector technology provides non-invasive X-ray-like scans to assess the stability of tailings dams, identifying material deterioration and water infiltration (Furlan, 2012; Swinburne University of Technology, 2023). • Nuclear Waste Monitoring: Monitoring muons to enhance data precision for nuclear waste storage and Dry Storage Containers (DSCs) at CANDU® facilities (Jonkmans et al., 2013). • Volcanic Activity Detection: Muon tomography combined with existing volcanic monitoring methods aids real-time detection of magma ascent, helping anticipate eruption magnitude and improve evacuation efforts (Leone et al., 2021). • CO₂ Geostorage Monitoring: Continuous, non-invasive muon tomography helps monitor CO₂ geostorage sites, ensuring secure storage and preventing leakage to mitigate climate change concerns (Gluyas et al., 2018).
	Application Muon detector technology offers opportunities for EduMOD by aligning with environmental monitoring needs (Furlan, 2012; Leone et al., 2021; Gluyas et al., 2018). Integrating this into EduMOD's STEM curriculum could enhance student engagement through real-world applications. However, the complexity and cost of the technology present a threat to its widespread adoption in schools with limited resources.

Figure 4. PESTLE Analysis

3.4. SWOT Analysis



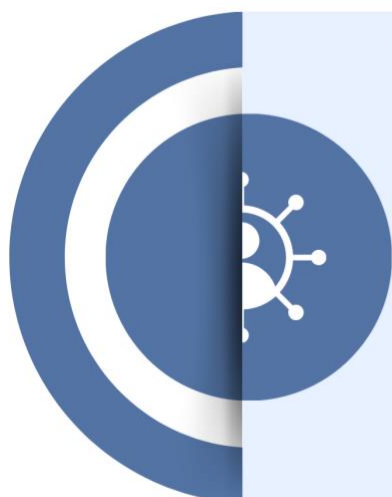
STRENGTHS

- mDetect's available reputation via word-of-mouth due to its current distributions in the United States lowers the barriers in customer acquisition efforts (Gildin, 2022).
- mDetect's technique leverages cosmic muons to generate X-ray-like scans of tailings dam structures, facilitating the identification of problems such as material deterioration or water infiltration. This technology enhances real-time monitoring capabilities, thereby advancing SDG 6 (Clean Water and Sanitation), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action), promoting sustainable and responsible mining operations (Swinburne University of Technology, 2023).
- mDetect's established in-house manufacturing and supplier process enables the firm to meet customer quantity demands without considering establishing potential subsidiaries in the target market.
- EduMOD provides a comprehensive package that includes curriculum alignment to the state syllabus, teacher resources such as lesson plans, and the ability to detect muons in the classroom, enabling practical applications of physics and mathematics, and teaching relativity through hands-on experiments (mDetect, 2025).



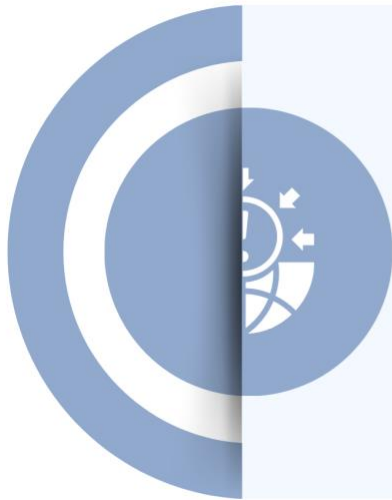
WEAKNESSES

- As mDetect relies on traditional marketing methods; it in turn has limited social media presence and is not able to effectively utilise the social media channel to attract potential customers, thus resulting in limited brand recognition (Pandey, 2021).
- Cheaper alternatives (CosmicWatch, UKRAA) as well as other indirect substitutes such as e-textbooks and conventional teaching techniques, provide a considerable obstacle to the customer acquisition process. These alternatives present reduced initial expenses, appealing to prospective clients and thereby intensifying competition for EduMOD in the educational sector (CosmicWatch, 2017; Rzyankina, 2024).
- mDetect's inability to develop an efficient distribution channel in the US, along with recent tariff regulations, results in considerable disruptions and logistical obstacles, constraining its ability to meet customer demands and capitalise on the burgeoning STEM education market in the U.S (Liu, 2025).



OPPORTUNITIES

- US Governments and educational institutions are significantly investing in STEM to equip students for the future. This corresponds with EduMOD's experiential learning methodology (Institute of Education Science, 2017).
- School districts, such as Palo Alto school district, has a very large budget proposition of \$8.6 million USD on instructional equipments and scientific textbooks (Palo Alto, 2024).
- The CA Science Framework 2016, which most school districts follow in California in 2025, has a specific focus on integration of science content and practical scientific application via experiments (California Department of Education, 2016).
- There is a large localisation opportunity in the US, as threats of suppliers are low due to the large scintillator market available in the US (IMARC Group, 2025).
- In the last twenty years, global discourse, rhetoric, recommendations, and policies concerning K-12 STEM education have intensified among businesses, industry, policymakers, think tanks, and educators. STEM education is regarded as essential for improving productivity, prosperity, and global competitiveness, while also addressing current and future socio-political, economic, and workforce challenges (Bryan & Guzey, 2020).



THREATS

- As previously mentioned, California and analogous areas are conducive to EdTech innovation. Startups may emulate or recreate analogous educational tools (Thomas & Nedeva, 2018).
- Although the increasing interest in educational technology, several schools remain in utilising textbooks owing to their cost-effectiveness and familiarity (López et al., 2021).
- Schools and universities may favor virtual physics labs and simulations (PhET, Labster) for three-dimensional learning which threatens to substitute the education sector's needs for the EduMOD (University of Colorado Boulder, 2019; Labster, 2019; Labster, 2024; Vergara et al., 2021).
- The budget proposition of prospective school districts such as Palo Alto appears to have a favoritism for scientific textbooks (Palo Alto, 2024).
- EduMOD's data sharing network (mNetwork) could conflict with strict US privacy laws like COPPA and FERPA, posing a challenge for compliance (Schifferle, 2020; Student Press Law Center, 2024).

Figure 5. SWOT analysis

4. Recommendation - Target Market

This section will focus on California as the primary segment, as it offers progressive curriculum standards such as the CASF and contains up to 66 high school districts in California (California Department of Education, 2024). The secondary segment, Texas, presents a long-term growth and therefore its market segment analysis will be relegated to [Appendix C](#).

4.1. Primary Market Segmentation

Using a multi-mode segmentation approach (geographic, demographic, psychographic, behavioural variables), the primary segment is as follows:

District-Level Decision Maker from High-Performing Public High Schools in California



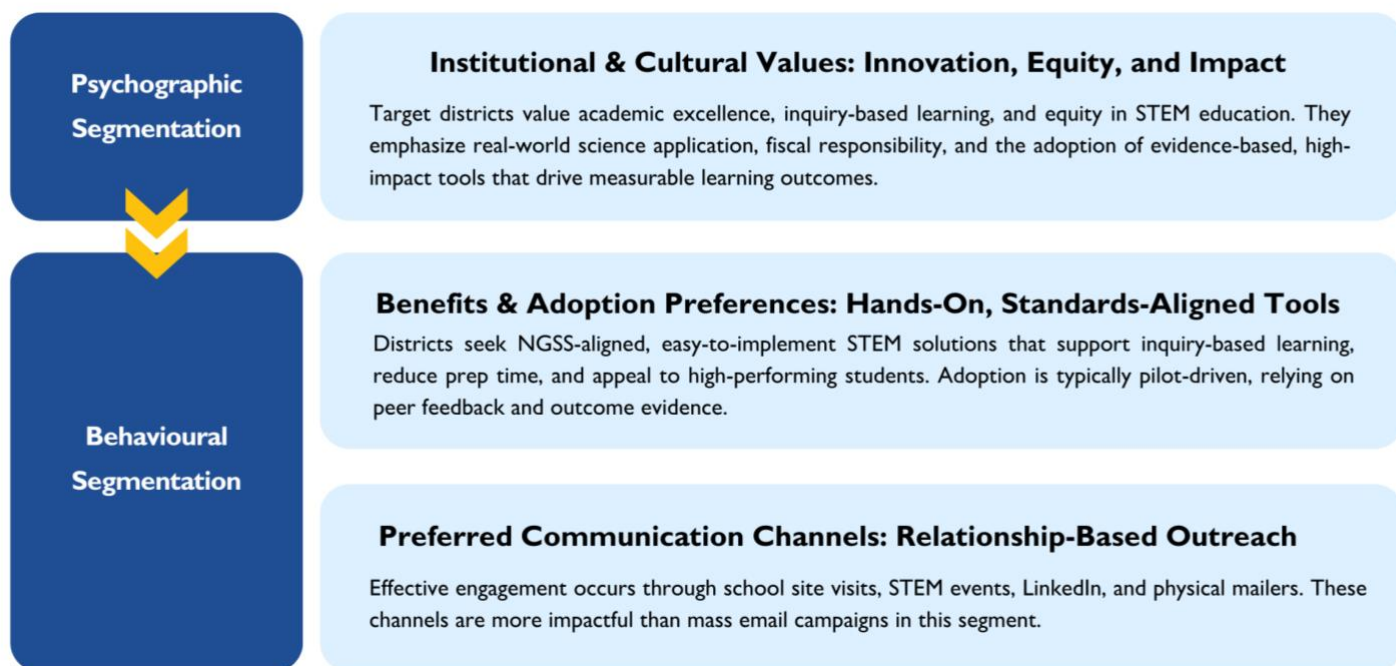


Figure 6. Target Market Segment

EduMOD's primary market entry should focus on district-level decision-makers in high-performing California public high schools, leaders empowered to drive early adoption. These directors oversee curriculum procurement and play key roles in STEM initiatives. Targeting innovation-driven districts allows EduMOD to validate its solution in real classrooms and build momentum for broader adoption.

To succeed, EduMOD must position itself as a plug-and-play, standards-aligned STEM solution that addresses core priorities: simplifying curriculum delivery, reducing teacher prep time, enhancing real-world scientific learning, and supporting measurable outcomes. Its pilot-first approach, supported by data-driven feedback loops, provides a low-risk, high-trust entry pathway tailored to the needs of forward-thinking districts.

Importantly, this segment also fits the Early Adopters stage in the Technology Adoption Life Cycle (**TALC**). Engaging these early champions will help EduMOD build credibility, collect testimonials, and lay the foundation for scalable, sustainable market growth (See [Appendix D](#) for the Value Proposition Canvas and [Appendix E](#) for TALC alignment.)

4.2. Customer Personas and Empathy Maps

Based on EduMOD's segmentation strategy, the primary customer persona is developed in Figure 7. This person plays a pivotal role in EduMOD's go-to-market approach, as he influences district strategy, budget decisions, and tool adoption. To secure their support, EduMOD must demonstrate measurable outcomes, align with NGSS and the California Science Framework, and ensure fiscal responsibility.



Robert H. Griffin
President, Board of Trustees

Robert is a long-serving school board trustee with deep roots in San Mateo. As President of the SMUHSD Board, he helps shape district-wide strategies and policies impacting over 8,000 high school students. With a professional background in accounting and governance, Robert brings a data-driven, equity-focused approach to school decision-making, prioritizing student achievement and instructional innovation.

Goals

- Ensure district-wide access to high-quality STEM resources
- Promote equity across all student demographics
- Support fiscally responsible innovation in classrooms
- Position SMUHSD as a leader in academic excellence and college readiness

Motivations

- Passionate about bridging opportunity gaps through education
- Believes in long-term ROI from investing in innovative tools
- Committed to preparing students for future-ready careers in STEM fields

Background

AGE: 50
OCCUPATION: President, Board of Trustees of San Mateo Union High School District
YEARS IN EDUCATION GOVERNANCE: 28+ Years
INTERESTS: Education equity, public governance, student achievement, STEM advocacy, compliance, budget oversight, instructional quality

Challenges

- Navigating multi-stakeholder consensus on new technology
- Balancing innovation with tight district budgets
- Ensuring new tools align with compliance and performance standards

Preferred Channels

Icons: Email, LinkedIn

Figure 7. Target Segment Persona

As shown in the empathy map (Figure 8), Robert balances pride in district innovation with caution toward unproven tools. He is motivated by long-term ROI and equity-driven outcomes but faces challenges such as stakeholder consensus, tight budgets, and proposals lacking data. EduMOD must address these concerns with streamlined implementation, clear classroom impact, and strong compliance alignment.

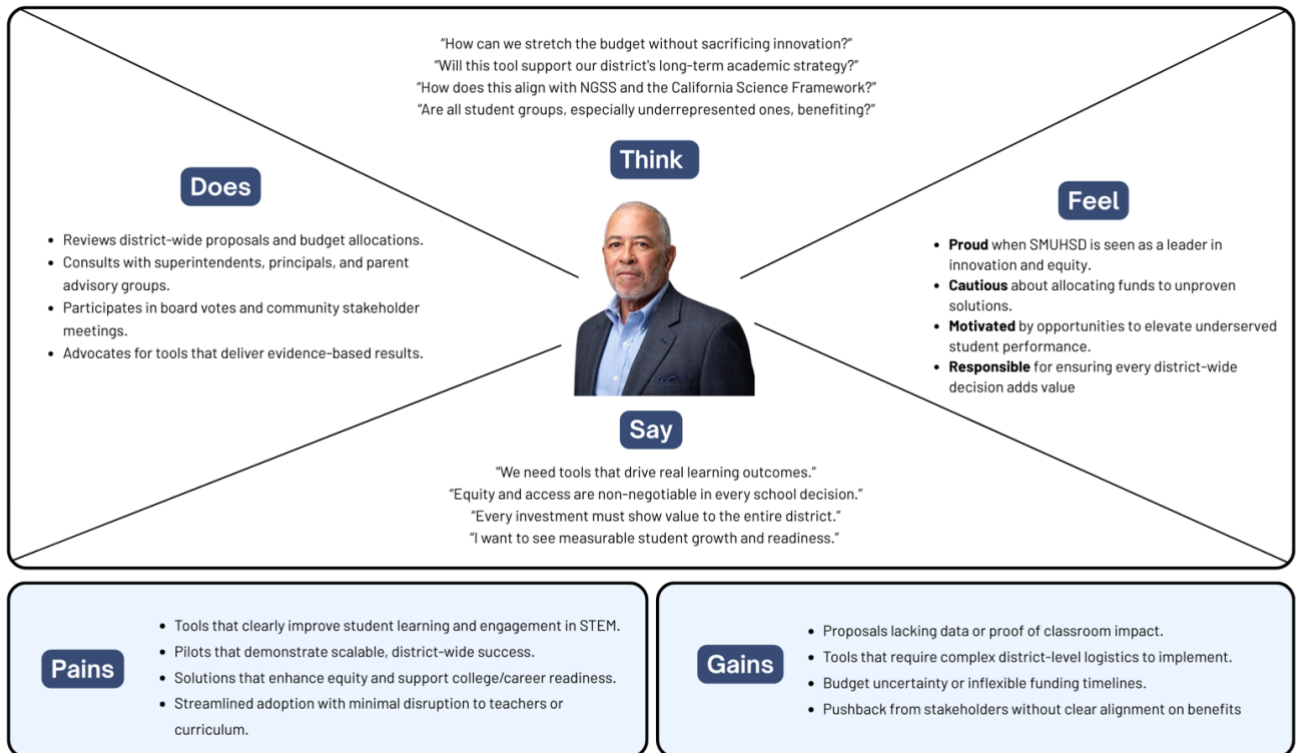


Figure 8. Target Segment Empathy Map.

5. Product Enhancement and Adaptation

5.1. Application Formation and Curriculum Enhancement

5.1.1. Application Development

Based on our SWOT analysis, an opportunity can be found in the significant investment of the US Government in STEM programs (Institute of Education Science, 2017). In recent years, the widespread adoption of smartphones with computing capabilities akin to desktop computers has led to a significant increase in the popularity of mobile applications (Wai et al., 2016).

Currently, EduMOD continues to use conventional techniques for distributing learning materials to instructors and learners, predominantly utilising platforms like Google Drive. This technique is constrained in its ability to offer interactive, hands-on experiences, which are essential for promoting deeper engagement and maintaining student interest.

To address this gap, it is recommended that mDetect consider the development of a dedicated educational application, boosting user engagement via interactive elements and providing a dynamic, immersive learning environment.

The application should include these features:

Features	Details
Fundamental Learning Engine and Content	Interactive Simulations: Users would not merely read about muon decay, time dilation, or detection techniques; they would engage with simulations. They may modify settings (such as muon velocity and detector material) and observe the outcomes visually. Visualise particle trajectories, decay phenomena, and relativistic effects. Guided, Bite-Sized Modules: Deconstruct intricate subjects (Standard Model framework, atmospheric muon generation, muon g-2 anomaly, muography applications) into digestible, interactive lessons that advance in complexity.

Motivation & Structure	Gamified Progression: Implement points (XP) for lesson completion, problem-solving, and interaction with simulations. Utilise tiers or unlocking modules (e.g., "Muon Fundamentals," "Relativistic Muons," "Muon Experiments," "Beyond the Standard Model"). Learning Streaks & Daily Goals: Promote consistent learning with daily streaks for completing brief lessons or challenges, hence facilitating habit formation.
Assessment & Practice	Varied Question Types: Incorporate brief multiple-choice and true-false assessments, with more intricate enquiries necessitating numerical responses, formula utilisation, diagrammatic representations (such as Feynman diagrams or basic detector illustrations), or concise written elucidations. Engaging Review Sessions: Periodically employ a rapid, Kahoot-style quiz format for an engaging and swift refresher of essential definitions, constants, or particle characteristics (Maraza-Quispe et al., 2024).
Targeted Feedback	Intelligent Feedback on Problem Solving: This is essential for physics. If a user makes an inaccurate calculation, the application should not merely indicate "incorrect." It should preferably evaluate the steps (if provided) or the nature of the problem to indicate potential sources of the mistake (e.g., "Did you consider relativistic effects?" or "Verify your unit conversion for energy").

Table 3. EduMOD application's features

The app's gamified features, such as achievement levels and XP points, would further motivate students to engage with the material, while its adaptive learning engine tailors' content to individual progress, ensuring effective and personalised education (Khaldi et al., 2023).

5.1.2. Curriculum alignment

California's geology and mining sectors are predicted to gain increasing strategic importance over the next decade. (California Department of Conservation, 2023). The California Energy Commission (2023) and Workforce Development Board (2023) emphasize the growing need for geoscience

professionals, environmental surveyors, and mining engineers to support sustainable development goals.

As such, it is recommended that mDetect expand mNetwork by introducing geology-related educational modules targeting high school senior science classes and career exploration programs.

Specifically, mDetect should develop career-linked projects that integrate physics (muon interaction with matter) and earth sciences (geological formations), aligning with both NGSS Earth and Space Science standards (Next Generation Science Standards, 2013) and California's CTE Standards for Energy, Environment, and Utilities Career Pathways (California Department of Education, 2022).

This strategy would significantly increase curriculum relevance, as Earth and Space Science has become a required dimension under the NGSS three-dimensional learning model (Next Generation Science Standards, 2013).

The strategy would also align EduMOD with California's workforce development priorities, which emphasize preparing students for future careers in renewable energy, resource management, and sustainability industries (California Workforce Development Board, 2023).

5.2. Pricing Strategy

Based on our SWOT analysis, EduMOD's main weakness is its price point concerning competitors such as CosmicWatch and UKRAA (CosmicWatch, 2017; Rzyankina, 2024). EduMOD is currently priced at AUD \$2,950. With a 15% international markup, it enters the U.S. at USD \$3,392.50 (mDetect, 2025). This margin covers shipping, setup, and localisation costs, consistent with standard international pricing practices (Bajpai, 2025).

5.2.1. Cost and break-even analysis

In Year 1, mDetect's projected annual operating cost in the U.S. is USD \$339,202 (refer to [Appendix E](#) for breakdown).¹ For calculation purposes, the cost of goods sold (**COGS**) per EduMOD unit is USD \$500, and mDetect expects to distribute 200 units, 170 through sales and 30 via leasing per annum.

Component	Value (USD)	Details
Annual Operating Cost	\$339,202	Includes staff, rent, equipment (Appendix 4)
COGS	\$500/unit	Material & manufacturing cost per EduMOD
Total EduMODs Distributed	200 units (170 sold, 30 leased)	Year 1 target
Required Total Revenue	\$424,202	Operating + product costs
Break-even Price/Unit	\$2,121	\$424,202 / 200 units

Table 4. Break-even and unit cost breakdown (Year 1)

mDetect must generate a minimum of \$424,202 in revenue to cover both operating costs and product expenses. This translates to a break-even point of USD \$2,121 per unit sold. Since the current unit price of USD \$3,392.50 exceeds this amount, mDetect maintains a gross margin of roughly 60%, providing strong financial leverage in its first year.

5.2.2. Recommended US Pricing Strategy

Using the breakeven data, mDetect's pricing model can be strategically structured to offer both direct sales and leasing options.

¹ These costs include staff salaries, office rental, office equipment, and a delivery vehicle, among others (Vidal, 2023).

Model	Price (USD)	Margin Over Cost	Notes
EduMOD Sale	\$3,392.50	~60% over \$2,121 breakeven	Retain current strategy
1-Year Lease	\$848/unit/year	~25% of sale price	Standard lease-to-buy ratio in edtech (Misha, 2024)
2-Year Lease	\$1,530 total	~10% discount	Encourages longer commitments
Lease-to-Own Option	Negotiated Buyout	Based on use/duration	Offers flexibility after 3-year lease

Table 5. Competitor Comparison and Analysis

To improve accessibility for U.S. schools:

Pricing Model	Details
Direct Sale	\$3,392.50 per unit (existing model)
1-Year Lease	Schools can lease an EduMOD for USD \$848 per year (discounted 25% from the sale price). This matches with the lease-to-buy ratios in the edtech sector, allowing schools to access advanced science modules without a large upfront investment (Misha, 2024).
2-Year Lease	A 10% discounted rate of USD \$1,530 for two years is recommended. This incentivizes longer commitments, reducing churn and simplifying revenue forecasting.
Lease-to-Own	After a 3-year lease, districts can purchase the units at a negotiated price based on usage. This model encourages longer-term relationships and is common in U.S. school procurement to manage capital expenditure risk (NCES, 2020).

Table 6. Pricing Model Strategy

This mirrors standard educational procurement behaviour and supports smaller school budgets (Misha, 2024; NCES, 2020).

5.2.3. Strategic Justification

The pricing strategy also fits seamlessly into the broader international market entry approach. mDetect will pursue a direct-to-district sales model via a California-based LLC. This not only retains pricing control but also enhances customer engagement and loyalty. Leasing options also align well

with U.S. school budget limitations, enabling wider adoption without compromising financial sustainability.

5.3. The Three Levels of Product

To reinforce its market position and align with the diverse needs of U.S. K–12 schools, a three-tiered product strategy has been developed to summarise the information analysed above. This approach ensures value delivery across functional, pedagogical, and strategic levels, responding directly to the curriculum alignment, budget constraints, and classroom engagement gaps identified in earlier sections.

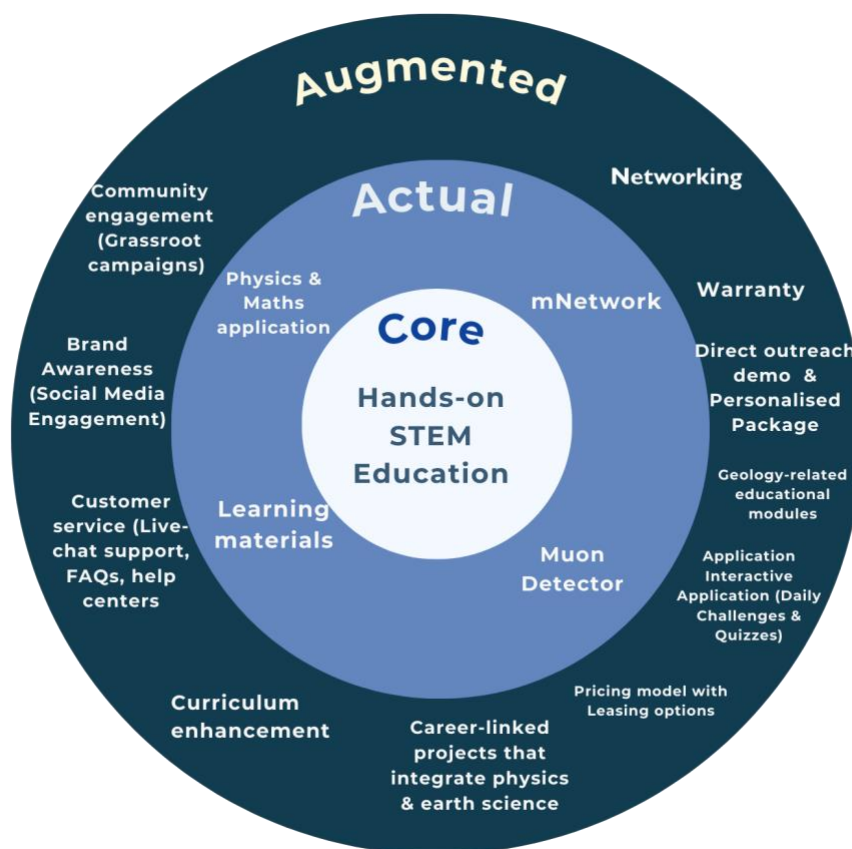


Figure 9. The Three Levels of Product for EduMOD enhancement

6. International Market Entry Strategy

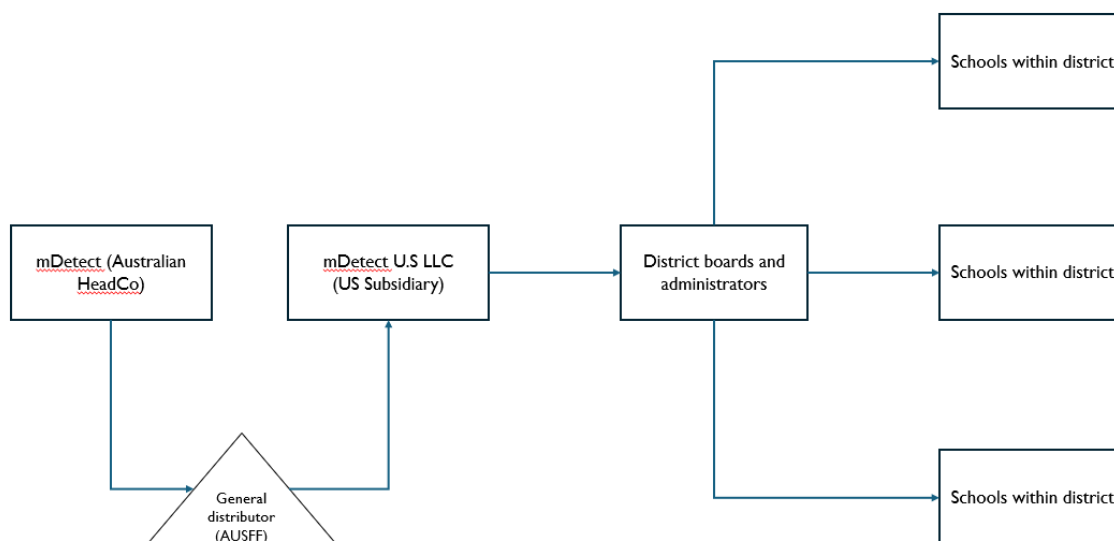
6.1. Entry Mode and Distribution

Based on our SWOT analysis, a major weakness that mDetect currently experiences in the United States is that it has a relatively weak distribution channel, reliant on specific education equipment distributors like PASCO Scientific.²

According to Diment (2025), there are only 552 school supplies wholesaling industries in the United States, only 15.3% of which (i.e 84 businesses) supply scientific equipment throughout the entire United States.

Therefore, we seek to recommend a direct sales entry mode into the market through a U.S subsidiary, which distributes to district administrators who would subsequently distribute them to the schools within the districts, illustrated as follows:

PROPOSED DISTRIBUTION NETWORK (USA)*



*Note: This distribution model is not concerned with communication and client acquisition methods, which uses a "ground-up" approach from individual teachers in order to reach district decision makers.

Figure 10. Proposed Distribution Network

² Refer to the Client Briefing Session

6.1.1. Subsidiary

The subsidiary is recommended as follows:

- An office space in San Jose to proximity with valuable school districts such as PAUSD (Palo Alto - Unified School District, 2024).
- Every model of manufactured EduMOD is to be imported into the USA through shipping partners such as AUSFF (AUSFF, 2025).
- The subsidiary should have 5 full-time office employees, 1 technician, and a casual delivery driver.

As informed by [Appendix F](#) and [Appendix G](#), the initial set-up costs for year 0 would be USD \$375,538. The estimated annual operating cost would be \$339,202 at year 1.³

6.1.2. 5-Year Growth Strategy

We give a generous estimate of a flat 15% growth in sales annually. We estimate that there would be an increase in distribution channels throughout other cities in California, such as San Francisco and Sacramento by the end of Year 5.⁴



Figure 11. Five-year growth strategy

³ This is based on the assumption that mDetect U.S LLC can distribute 5 EduMODS to 40 schools around various districts in California.

⁴ We project that mDetect U.S LLC would have a projected sales of 200 units of EduMODS (30 of which would be leased) 40 schools within various districts based on the assumption that each school requires 5 units (High-schools, 2024),

6.1.3. Estimated Financial Performance

Based on an estimated income statement ([Appendix H](#)), the profit margin of the U.S subsidiary is going to be -10.9% by the first year, then grows to 14.79% at Year 3 and 34.32% at Year 5. According to Seilier (2025), the U.S subsidiary would be able to outperform the current profit margin of the education sector at 6.7%.

According to our estimated balance sheet ([Appendix H](#)), the average fixed assets turnover ratio would be 12.93. As a good fixed asset turnover ratio is 5 (Remesh, 2024), the US subsidiary would be able to effectively utilise its investments to generate sales.

According to our cost-volume-profit analysis ([Appendix H](#)), it would take 211 units of EduMOD to break even. This means that for the first year, mDetect would experience a loss which gradually recovers at Year 3 due to growing sales.

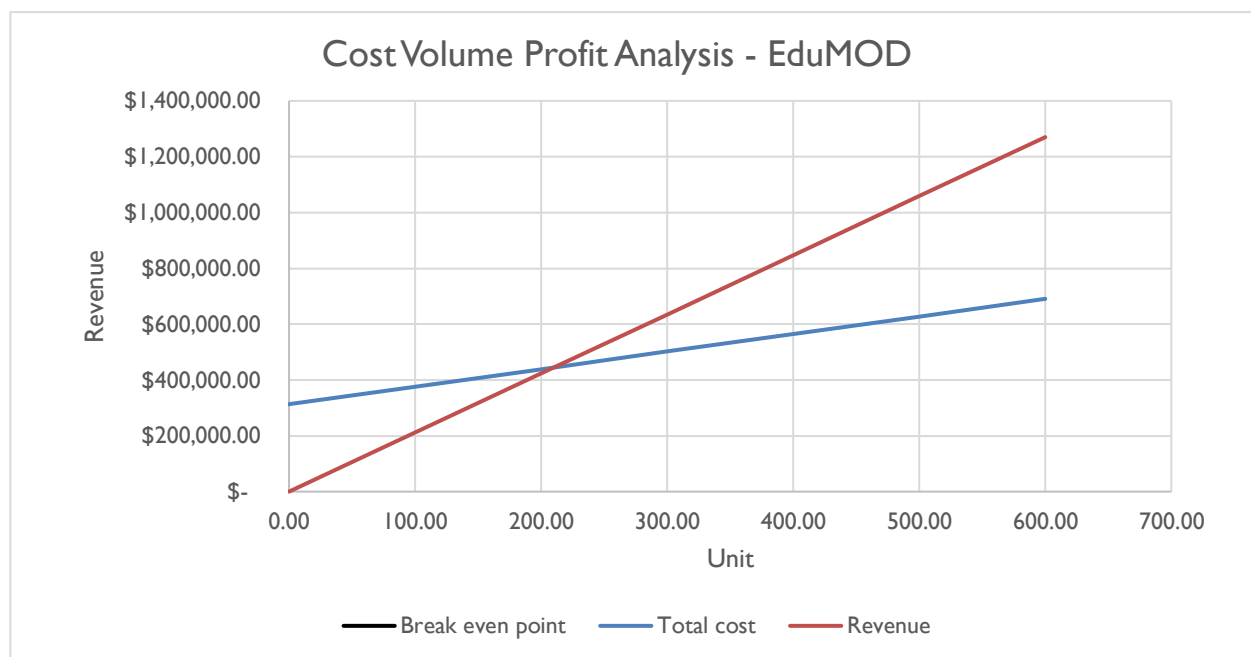


Figure 12. Cost Volume Profit Analysis

6.2. Value Proposition

At the core of EduMOD's international expansion is a strong commitment to educational impact. Its goals and values are as follows:

EduMOD Goals	EduMOD Values
- To transform STEM education through interactive, inquiry-driven experiences. - To cultivate scientific thinking and problem-solving skills in secondary students. - To promote accessibility, equity, and enthusiasm for advanced science topics.	Equity: Equal access to quality science tools, including students with disabilities. Innovation: Cutting-edge scientific exploration tools brought into classrooms. Collaboration: Building a global network of students and teachers through mNetwork. Practicality: Direct application of theoretical science via real-world data.

Table 7. EduMOD Goals and Values

To effectively position EduMOD in the U.S. market, it is essential to tailor its value proposition to resonate with the needs and priorities of its primary target segment according to Table 8 below:

Primary Segmentation	Value Proposition Statement	Alignment with EduMOD Goals and Values
District-Level Decision Makers in High-Performing California School Districts	As a district-level education leader, I see value in STEM tools that engage students with real-world scientific concepts through interactive, hands-on experiences, while being easy to integrate, scalable across schools, and supportive of our district's commitment to academic excellence, equity, and future-readiness	EduMOD directly aligns with this vision by offering a curriculum-enhancing, plug-and-play STEM solution that delivers interactive simulations, modular learning pathways, and gamified assessment tools. It supports equity through accessibility-focused design and offers flexible pricing models to accommodate district budgets. EduMOD empowers schools to deliver high-impact science education while minimizing teacher prep time and administrative complexity

Table 8. Value Proposition Statement

6.3. Branding and Communication Strategy

6.3.1. Brand Positioning

Positioning Statment



EduMOD offers a differentiated, hands-on STEM learning experience designed to help students explore advanced physics concepts through the real-time detection of cosmic rays using muon detectors. The solution enhances classroom engagement and deepens conceptual understanding in particle physics, relativity, and statistics while ensuring full alignment with STEM Education Standards

Positioning Map

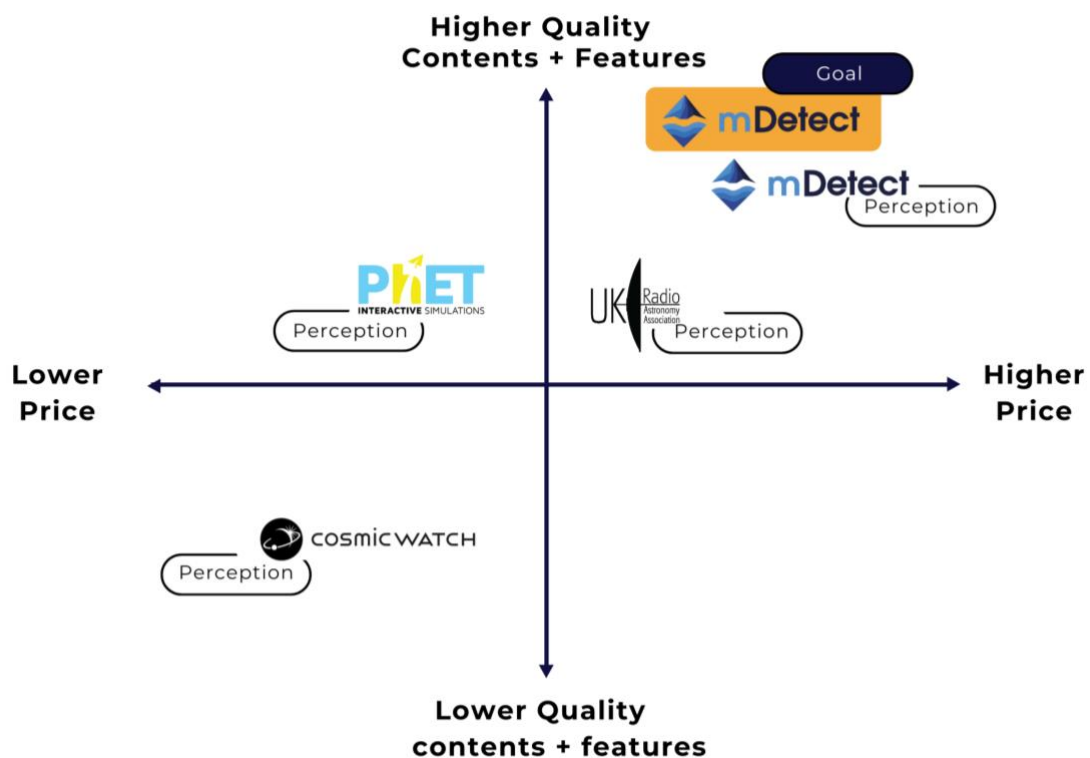


Figure 13. Positioning Map

This updated **positioning map** is developed from a perceptual map analysis comparing EduMOD with key competitors in the STEM education space to reflect mDetect's strategic direction for EduMOD based on its refined product and pricing strategies. While EduMOD is currently seen as a high-quality, high-cost solution, the **goal position** is to further enhance perceived educational value through a dedicated app, gamified modules, and Science integration, **without increasing price perception**.

The updated position shows a **vertical shift upward** to represent added content quality and curriculum alignment, while the **price level remains steady**, reflecting mDetect's new **tiered pricing strategy**. This repositioning ensures that EduMOD maintains its premium identity while offering **greater value at the same price**, enabling stronger adoption and long-term classroom impact.

All of the strategies presented below are meant to both spark interest and support from grassroots organisations (teachers, students, parents) and to establish the basis to persuade district-level decision makers such as Mr. Robert H. Gryphon, who represents EduMOD's core customer group in the US market.

6.3.2. Marketing Strategy

a) Email Marketing

Email marketing will be a cornerstone of EduMOD's outreach strategy, replacing mDetect's low-yield cold calling, which achieves just a 2.3% success rate (Cognism, 2025). Email campaigns deliver a \$42 return for every \$1 spent (MacDonald, 2019), with segmented efforts boosting revenue by up to 760%.

EduMOD will deploy a strategy focused on segmentation, personalisation, and trigger-based automation to maximise engagement and conversions. As shown in Figure 14, triggered campaigns that combine marketing and transactional content can achieve up to a 60% open rate, far exceeding direct (20%) or standalone transactional (52%) emails (MacDonald, 2019).

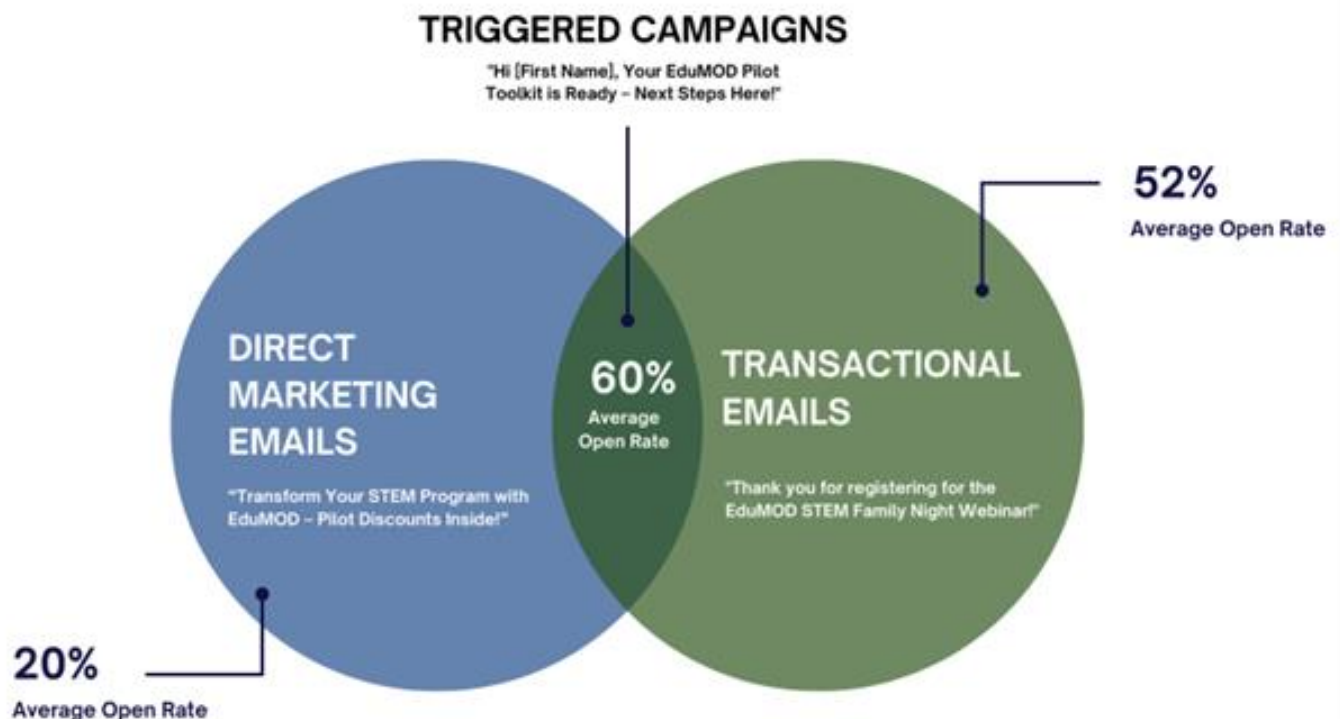


Figure 14. Effectiveness of triggered campaign

a) Social Media Marketing

EduMOD's social media strategy is central to establishing its reputation as a credible innovator in STEM education. By combining professional authority with authentic storytelling, the strategy will engage key stakeholders, STEM curriculum leaders, principals, district decision-makers, teachers, students, and parents, across school communities.

Leveraging a multi-platform approach, EduMOD will target both top-down decision-makers and grassroots influencers to build visibility, trust, and adoption momentum. This is supported by research showing that educational content increases consumer purchase likelihood by 131% (Cote, 2024). Content tone and format will be tailored to each platform to maximise impact across the K–12 STEM landscape.

A detailed plan for the social media platform is provided in [Appendix I](#), including LinkedIn for professional positioning, X (Twitter) for real-time updates, YouTube for video-based engagement, Instagram for community storytelling, and Facebook for outreach to PTAs and local school groups.

6.3.3. Onboarding Strategy

a) Direct Outreach Demo

The appointment of a “US Education Partnerships Manager” in San Jose is a key strategy for mDetect, to establish effective communication channels at the district level in the US education system. According to Solberg (2000), “embedded market knowledge” from a local representative can reduce the conflict between global standardisation and local needs, while ensuring consistency of brand messaging.

In the context of the US public education market being particularly fragmented and lacking in transparency, locating the right people with authority is a major challenge for vendors. Only 12% of finance personnel in public schools have a clear understanding of the procurement process and who the decision makers are (Fittes, 2024a; 2024b). At the same time, only about 30% of principals have full authority to approve equipment purchases without seeking approval from their superiors (Fittes, 2025). These figures clearly show that a local representative is a key factor in shortening decision cycles and improving conversion rates.

b) Handwritten Letter and Personalised Package Strategy

As a core part of its personalised outreach strategy, EduMOD should implement handwritten letters and printed materials to key school personnel, including principals, science department heads, and Parent Teacher Association (PTA) representatives. The letters will also include EduMOD brochure printed using offset printing on premium paper, and a flyer presenting successful pilot data from innovative districts such as PAUSD and MVLA.

Sending handwritten letters and personalised materials in conjunction with an email marketing campaign was essential to increasing message recall and building brand recognition with potential customers. According to Wufoo research, sending handwritten letters increased customer retention by up to 50% compared to not sending them (Lentz, 2024).

The implementation strategy was divided into two phases to optimise outreach and increase conversion rates. Specific information on implementation timing and target audience is presented in [Appendix J](#).

c) Grassroots Advocacy Campaign

PTAs do not directly make purchasing decisions, but they play an important role in shaping school priorities and influencing board-level decisions. PTAs have funded instructional materials and provided funding for additional educational staff (Dharmaraj et al., 2017).

EduMOD will partner with local PTAs to organise a series of “STEM Family Night” events, interactive science experiences led by students. These sessions will include hands-on experiments with EduMOD devices and feedback from parents and teachers on the impact of the product during the pilot phase. In parallel, EduMOD will release a “STEM at Home” activity kit for families, inspired by PTA initiatives and the national STEM movement.

7. Evaluation of Recommendations

	Suitability	Acceptability	Feasibility
Establishing U.S Subsidiary to distribute to nearby school districts	Appropriately counteracts small scale of school supplies wholesaling industry in the US (Diment, 2025). As there is a disbursement of 179.1 billion USD to state and education agencies in 2024, establishing the U.S subsidiary is highly suitable as most school districts would receive additional fundings (U.S Department of Education, 2024).	The initial outlay and investments might concern the investors and shareholders of mDetect, However, the positive outlook and growth of the public education sector in California would be able to justify long-term benefits and operational success (Murphy & Paluch, 2018).	Establishing a subsidiary in the United States would require a considerable amount of capital and operational expenses (i.e \$339k) within the first year. However, as mDetect can raise a significant amount of budget, the significant initial investment can be justified with high return on investments within 5 years of operations. (See Appendix H)
Pricing Strategies	The pricing strategy supports mDetect's international expansion goals while maintaining global competitiveness (Bajpai, 2025). By aligning with common international markup practices and agreement with EduMod's team, such as a 15% markup used here, and adopting a direct-to-district model, it	The set unit price of USD \$3,392.50 yield a gross margin of ~60%, making the financial performance acceptable to CEO and key stakeholders (Vidal, 2023). The tiered leasing options, \$848/year and \$1,530/2 years - reduces purchasing barriers for budget-constrained U.S. schools and districts; making the pricing strategy easy to recommend to key	The current cost model, based on a COGS of USD \$500 per unit and projected Year 1 operating costs of USD \$339,202, is financially viable (Appendix F & Appendix G). mDetect is forecasted to generate over \$424,000 in revenue, covering costs and allowing early-stage self-sufficiency despite initial overheads (Vidal, 2023).

	ensures pricing control and strong alignment with localized procurement behavior in the U.S. (Diment, 2025; PAUSD, 2024).	stakeholders (Misha, 2024; NCES, 2020).	
Marketing and onboarding Strategies	Aligns with EduMOD's goal of reaching decision makers at the school and district levels in the U.S. education system through a local representative (Solberg, 2000). Personalized letters and packages increased message retention amid information overload, and advocacy through PTA and student voice created organic demand from the community. (Lentz, 2024).	Initial investments to deploy U.S. representatives, produce specialised mailers, and coordinate community events raises operational costs and overheads (See Appendix G) The growing importance of STEM education, the growing need for community engagement in educational decisions, and successful models from initiatives such as National PTA's STEM+Families strongly support the acceptability of these strategies (Dharmaraj & et al., 2017).	The high open and response rates recorded from the handwritten letters (99% open rate, 30% response rate) (Stein, 2021), coupled with the moderate scale of PTA coordination, made the implementation process practical and resource efficient.
Develop an application and curriculum development	The application's emphasis on immersive STEM education via interactive muon detection simulations meets the rising demand for engaging, hands-on learning	The application is expected to be positively received by the target demographic, including students, educators, and academic institutions. The use of gamified elements would markedly improve student	The incorporation of interactive simulations, real-time data processing, and adaptive learning capabilities is technically achievable with contemporary software development tools and

for learning materials	experiences (Spector, 2024). Developing geology-focused modules on MNetwork aligns with mDetect's strategic objective to expand its educational footprint and diversify the academic applications of muon detection (California Workforce Development Board, 2023).	engagement by incentivizing regular involvement with the material (Lampropoulos & Kinshuk, 2024). The proposed content is likely to be well-received, particularly by senior science classes and CTE-aligned programs (Next Generation Science Standards, 2013; California Department of Education, 2022).	cloud-based platforms (mDetect, 2025; Fabian et al., 2015). The initial overhead on development, however, may represent funding challenges for mDetect and investors. Adapting the platform to deliver earth science content would require collaboration with curriculum experts and additional testing to ensure alignment with both NGSS and classroom implementation standards.
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Table 9. Evaluation of Recommendations

8. Conclusion

In providing recommendations for mDetect's international expansion of the EduMOD, we have chosen the United States market due to its large funding pool and opportunities for new entrants like the EduMOD into the STEM sector.

We have identified that the main problems impacting EduMOD's entry into the US are four-fold: inefficient pricing strategy, limited social media presence, and a bottleneck in distribution channel due to the small size of the school supply wholesaling industry in the US.

Our recommendations counter these problems simultaneously through the establishment of a subsidiary in San Jose and reach the district board level through a combination of handwritten letters, personal presentations and grassroots advocacy campaign. The pricing strategy involves multiple leasing options on top of direct sales. We also recommend for an adjustment of educational applications and curriculum adaptations for geology modules in the US.

We project that through this recommendation, the EduMOD would be able to break even and turn profits at Year 3 and would be able to reach school districts in San Francisco based on our sales growth estimate and marketing strategy.

This project is in line with the EduMOD's value proposition by targeting a market that has a specific focus on alignment with the NGSS and practical application of STEM. However, as schools may still favor virtual physics lab and traditional textbooks, mDetect may face challenges in increasing engagements with school districts.

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Appendices

Appendix A. Competitor Strengths

Brands	Strengths
Cosmic Watch	CosmicWatch, developed by MIT students, is an affordable and user-friendly muon detection device designed for students and educational institutions, hence expanding its accessibility to a wider and cost-conscious market. It comprises a self-sufficient device utilising a plastic scintillator and silicon photomultiplier for light collection, facilitating ease of operation. The apparatus is powered by batteries and operates with the included software to facilitate interactive physics measurements. At a price of \$100, CosmicWatch serves as an economical and accessible resource for educational institutions, providing a budget-friendly tool to investigate essential physics concepts such as cosmic rays. Its simplicity, user-friendliness, and affordability (a domain where EduMOD currently has a shortcoming) render it an outstanding competitor in the educational technology sector.
UK Radio Astronomy Association (UKRAA)	The UK Radio Astronomy Association (UKRAA)'s muon detector is marketed as a medium-high quality product; however, its restricted availability and high price augment its appeal as an exclusive product. As a non-profit entity, UKRAA mainly focuses on scientific progress and quality over commercial pursuits, rendering the detector especially appealing to universities and research laboratories. This strategic approach highlights the detector's significance in innovative research and attracts customers dedicated to progressing science with sophisticated and reliable technology.
PhET	PhET operates in the mid-tier market for quality, providing an extensive array of STEM activities beyond mere physics, and at a substantially cheaper price point than EduMOD, with numerous materials accessible almost at no cost. PhET's affordable cost makes it an attractive choice for consumers prioritising theoretical learning, such as assignment completion and exam preparation, over hands-on, interactive experiences. This stance enables PhET to target a small group of consumers interested in affordable STEM education while prioritising theoretical over practical, real-world applications. Unlike EduMOD, which focuses on immersive, experiential learning, PhET offers accessible online resources for individuals pursuing a more economical method of STEM education, without emphasising physical experimentation.

Table 10. Explanation for Competitor Strength

Appendix B. Explanation for Brand Position

Brands	Explanation for Position
EduMOD	EduMOD is positioned in the upper-right quadrant, indicating high-quality content and features , and a higher price . With its wide-ranging packages that includes physics learning materials and a global sharing network, EduMOD appeals to customers who are willing to pay a premium for high-quality and STEM-related content. This aligns with mDetect's strategies that enable students to explore science topics through hands-on experimentation.
CosmicWatch	CosmicWatch is situated in the lower-left quadrant, representing a lower pricing point and moderate quality of content and features . This strategic position indicates that CosmicWatch may predominantly depend on cost-effectiveness and branding to appeal to clients seeking STEM educational tools, without a strong emphasis on the profundity of learning materials or sophisticated functionalities. The offering is expected to attract those prioritising affordability in a STEM solution, where cost outweighs the importance of content quality or customer service.
UK Radio Astronomy Association (UKRAA)	Similar to EduMOD, UKRAA's product represents the premium market, providing medium-high category features at a higher price range . The brand underscores scientific progress and ensures restricted availability, attracting affluent, science-oriented clientele, especially individuals from academic institutions and research facilities. This positioning aims at institutions and individuals who value advanced technology and are prepared to invest in products that enhance scientific research and development.
PhET	PhET is situated close to CosmicWatch, providing moderate quality at a low price . This positioning appeals to budget-conscious consumers interested in STEM educational tools, yet who do not emphasise experiential learning opportunities. In contrast to EduMOD, PhET is not a direct competitor, as it predominantly emphasises online platform-based learning, rendering it more appropriate for users primarily interested in theoretical content and assignments, rather than interactive, practical implementations of STEM principles.

Table 11. Explanation for Competitor Landscape Position

Appendix C. Secondary Market Segment

School-level Decision Maker from STEM-Focused Magnet High Schools in Texas

Segmentation Category	Variable	Details
Geographic Segmentation	State	Texas, the second-largest K–12 market (5.84 million students). Emphasises career readiness, STEM integration, and curriculum alignment via TEKS standards.
	School District & Magnet School	EduMOD will target STEM-focused magnet high schools in major metropolitan areas such as Dallas, Austin, and Houston. These public institutions offer specialized tracks in science, technology, and engineering, attracting high-performing students (Magnet Schools of America, 2025). Additionally, these schools comply with IDEA, Section 504, and the ADA, ensuring equitable access for students with disabilities, aligning with EduMOD’s commitment to inclusive STEM education.
Demographic Segmentation	Academic Level & Program Type	Focus on Grades 9–12, specifically students enrolled in college preparatory STEM pathways, AP science courses, and project-based learning environments. These magnet schools often cater to academically advanced, university-bound learners.
	Funding Environment	Schools can access STEM-specific grants, federal programs, and technology adoption funding to support specialised learning tools and curriculum-enhancing technology.
Psychographic Segmentation	Institutional Priorities	Magnet high schools emphasise academic excellence, research-driven learning, and equitable access to high-impact science and technology experiences. Educators are motivated to adopt tools that mirror real-world scientific applications and build long-term student interest in STEM fields.
	Leadership Preferences	School leadership prioritises tools that align with state standards, improve college and career readiness, and include professional development for effective implementation. Leaders seek evidence of instructional impact and ease of classroom integration.
Behavioral Segmentation	Benefit Soughts	Educators and district decision-makers seek curriculum-aligned, interactive STEM tools that promote active student engagement, foster hands-on experimentation, and support inquiry-based learning in classroom settings.

	State Alignment	The product must align with the Texas Essential Knowledge and Skills (TEKS) standards to be considered for classroom use. TEKS compliance ensures instructional relevance and curriculum integration (Texas Education Agency, 2023)
	Compliance Requirement	Tools must meet student data privacy regulations, including FERPA and COPPA. Additional adoption enablers include teacher training support, bilingual accessibility, and the ability to scale across multiple campuses.

Table 12. Secondary Market Segment

Value Proposition Canvas

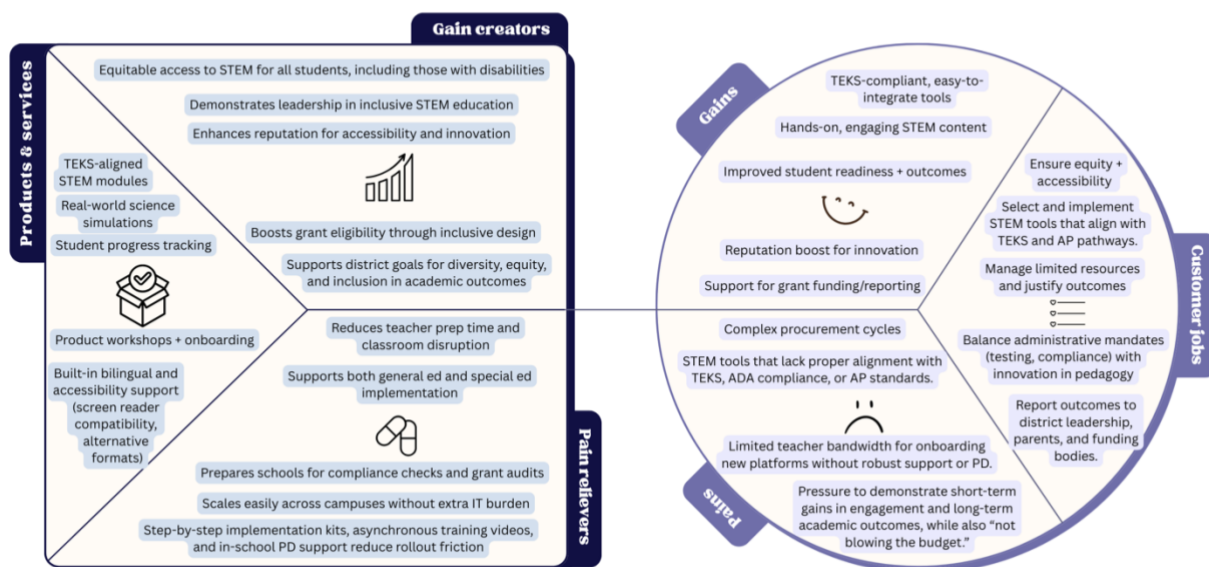


Figure 15. Value Proposition Canvas of Secondary Market Segment



Figure 16. Secondary Market Segment Persona

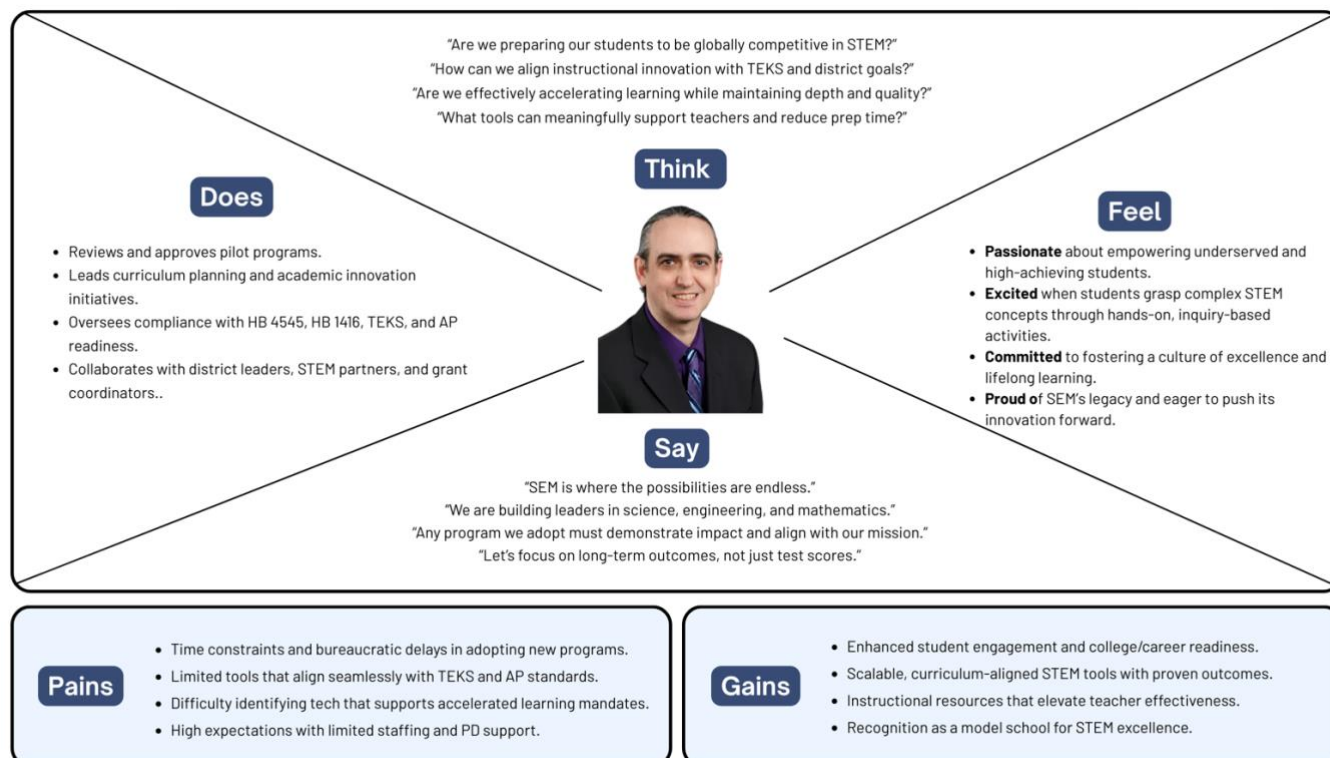


Figure 17. Secondary Market Segment Empathy Map

Appendix D. Value Proposition Canvas

The Value Proposition Canvas (Figure 18) outlines how EduMOD directly addresses the needs of its primary market segment: district-level decision makers from high-performing public high schools in California. These schools seek NGSS-aligned, hands-on STEM tools that are easy to implement, cost-effective, and enhance student engagement.

EduMOD's offering, such as its plug-and-play muon detector, mNetwork collaboration platform, and curriculum-ready lesson plans, solves key pain points like prep time, procurement barriers, and curriculum integration. By aligning with the segment's priorities and constraints, the canvas reinforces EduMOD's strong product-market fit and guides the internal market entry strategy

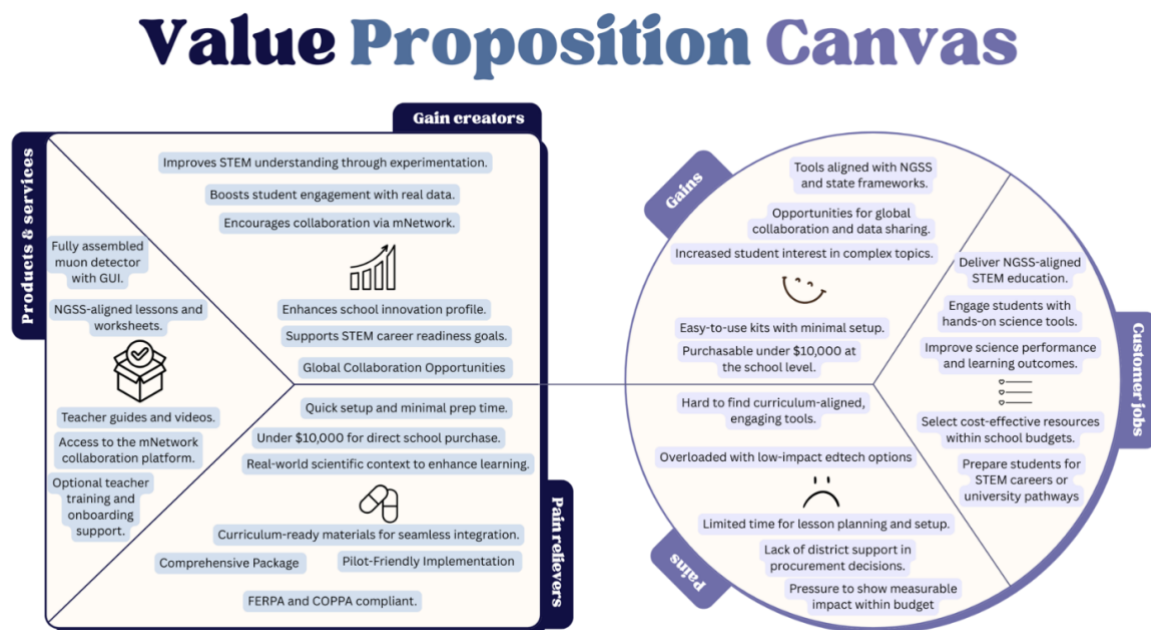


Figure 18. Value Proposition Canvas

Appendix E. The Product Adoption Curve

Understanding the five stages of the Technology Adoption Life Cycle (**TALC**) is essential for positioning EduMOD in the U.S. education market (Stiris, 2024). As a first-entry product in the U.S., EduMOD targets the early market, specifically the 13.5% of users classified as Early Adopters (Figure 19). These include forward-thinking schools in high-performing California districts like Palo Alto Unified, San Mateo Union, and Cupertino Union. They are ideal because they actively seek innovative tools that enhance learning outcomes and are willing to adopt new technologies early, even before widespread market validation.

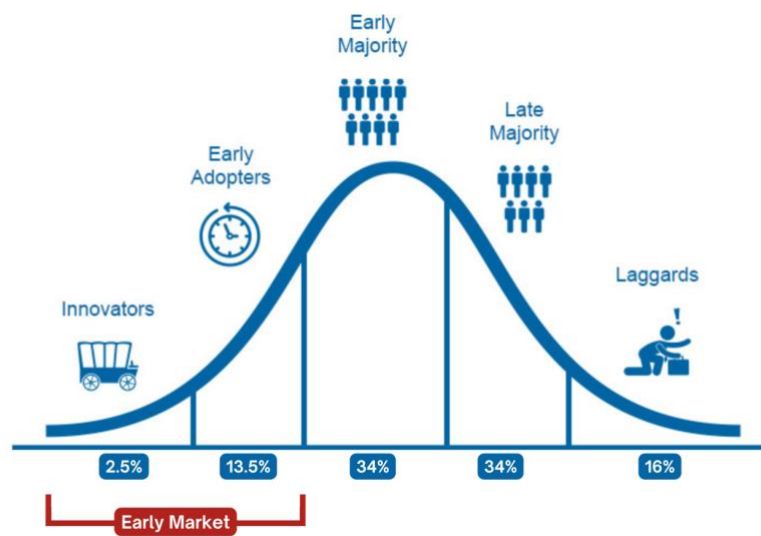


Figure 19. 5 Stages of New Technology Adoption

Aligning with Early Adopters

This group is open to pilot trials and early-stage tools, provided there is alignment with curriculum standards and a clear instructional benefit (Stiris, 2024). They are driven by academic differentiation and actively participate in STEM events, grant programs, and district innovation initiatives, seeking solutions that elevate their school's STEM profile.

Challenges in the Adoption Journey

Despite their openness, Early Adopters still require validation through case studies, testimonials, and measurable outcomes. They expect easy classroom integration, proof of compliance (e.g., FERPA, COPPA), and must plan around tight academic calendars. Addressing these needs is critical for timely adoption and securing district-level buy-in.

Appendix F. Set up costs in San Jose

Office cost

- Location: 1580 Oakland Rd (LoopNet, 2025)
- Area size: 1,811 Square Foot.
- Cost per Square Foot: \$19.20/ Square Foot / Year
- Annual lease rate: USD \$34,771.2



Figure 20. Office Space at 1580 Oakland Rd (San Jose, CA)

Set-up Capital

Item	Costs (USD)	Details & Source
Article of Organisation	\$70	(California Secretary of State , 2022)
Office furniture	\$1,899.9	Desk: 279.99/desk (IKEA, 2024) Swivel chair: \$99.99/chair (IKEA, 2024). Costs accounted for 5 office employees
Office Equipment	\$5,324.85	Monitors: \$74.99/ monitor (BestBuy, 2025). Docking Station: \$139.99/ docking station (BestBuy, 2025). ThinkPad laptop: \$849.99/Laptop (BestBuy, 2025). Costs accounted for 5 office employees.
Delivery car	\$35,389	(Autotrader, 2022)
Total	\$42,683.75	

Table 13. Set-up Capital Breakdown

Appendix G. Estimated operating costs in San Jose

Item	Cost	Source and Basis
Manager of Operations	\$70,000	(LinkedIn, 2025)
Brand Representative	\$48,000	(LinkedIn, 2025)
Service Manager	\$70,000	(LinkedIn, 2025)
IT Technician	\$50,000	(LinkedIn, 2025)
Bookkeeper	\$20,000	Part-time employment of bookkeeper to log invoices (LinkedIn, 2025).
Business Agent	\$49	(California Registered Agent Services, 2025)
Annual tax compliance service	\$280	(Block Advisors, 2023)
Electricity	\$11,900	(Pham, 2023)
EduMOD import costs	\$25,800	There are 82 highschools in San Jose (High-schools, 2024). Assume that each school would receive about 5 EduMODs and mDetect is able to distribute to 40 schools within various school districts. Importation is estimated to be 202 AUD (129 USD) per EduMOD (AUSFF, 2025). Import cost: $129 \times 5 \times 40 = \$25,800$
Internet	\$1,800	Mid-range price for AT&T Fibre at \$150/month (Supan, 2025).
Office Rent	\$34,771.2	From Appendix F
Software cost	\$534	Microsoft office 365 Enterprise: \$12 USD/month (Microsoft, 2025). QuickBooks: 32.5/ month (QuickBooks, 2024).
Depreciation	\$6,068	Assumed flat line depreciation over 7 years for all assets
Total	\$339,202	

Table 14. Operating Cost per year

Appendix H. Income Statement, Balance Sheet and Cost-Volume Profit Analysis

Our financial statements and performance metrics are made based on the assumption that mDetect U.S LLC can distribute 170 units and lease 30 units of EduMODs within the first year, growing 10% per annum. The cost of goods sold for each product is presumed to be \$500 per unit.

mDetect U.S LLC - Income Statement (Per annum)						
	Year 1	Year 2	Year 3	Year 4	Year 5	
Sales Revenue	\$ 368,331	\$ 423,580	\$ 487,117	\$ 560,185	\$ 644,212	
Leasing income	\$ 13,000	\$ 14,950	\$ 17,192	\$ 19,771	\$ 22,737	
less COGS	\$ 80,000	\$ 92,000	\$ 105,800	\$ 121,670	\$ 139,921	
Gross profit	\$ 301,330	\$ 346,530	\$ 398,509	\$ 458,286	\$ 527,029	
Less Expenses						
Set up costs	\$ -	\$ -	\$ -	\$ -	\$ -	
Employees	\$ 258,000	\$ 258,000	\$ 258,000	\$ 258,000	\$ 258,000	
Annual tax compliance service	\$ 280	\$ 280	\$ 280	\$ 280	\$ 280	
Registered business agent	\$ 49	\$ 49	\$ 49	\$ 49	\$ 49	
Electricity	\$ 11,900	\$ 11,900	\$ 11,900	\$ 11,900	\$ 11,900	
Import costs	\$ 25,800	\$ 28,380	\$ 31,218	\$ 34,340	\$ 37,774	
Internet	\$ 1,800	\$ 1,800	\$ 1,800	\$ 1,800	\$ 1,800	
Office Rent	\$ 34,771	\$ 34,771	\$ 34,771	\$ 34,771	\$ 34,771	
Software cost	\$ 534	\$ 534	\$ 534	\$ 534	\$ 534	
Depreciation	\$ 1,032	\$ 1,032	\$ 1,032	\$ 1,032	\$ 1,032	
Total expense	\$ 334,166	\$ 336,746	\$ 339,584	\$ 342,706	\$ 346,140	
NET PROFIT BEFORE TAX AND DEPRECIATION	-\$ 32,836	\$ 9,784	\$ 58,925	\$ 115,580	\$ 180,889	
Tax on Net Income	\$ -	\$ -	-\$ 12,374	-\$ 24,272	-\$ 37,987	
NET PROFIT AFTER TAX	-\$ 32,836	\$ 9,784	\$ 71,299	\$ 139,852	\$ 218,875	
Assumptions:						
(1) Depreciation at 7 years for furnitures and equipments						
(2): Annual sales volume assumed to be 170 units at year 1, assume leasing income over 5 years for 30 units.						
Flat-line growth of 15% assumed p.a						
This is based on assumed first year sale volume of 200 units						
(3): Assumed cost of goods sold is \$400 per unit						
Ratios	Year 1	Year 2	Year 3	Year 4	Year 5	
ROI	-9.826%	2.905%	17.352%	33.726%	52.259%	
Profit Margin	-10.90%	2.82%	14.79%	25.22%	34.32%	

Table 15. Income Statement per year

mDetect U.S LLC - Balance Sheet (Per annum)						
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Current Asset						
Cash	\$ 333,014	\$ 333,014	\$ 275,143	\$ 256,891	\$ 284,330	\$ 364,457
Profit/Loss for Year	\$ -	\$ 57,871	\$ 18,252	\$ 27,440	\$ 80,127	\$ 140,873
Closing Balance Cash	\$ 333,014	\$ 275,143	\$ 256,891	\$ 284,330	\$ 364,457	\$ 505,330
Fixed asset						
Furniture	\$ 1,900	\$ 1,900	\$ 1,900	\$ 1,900	\$ 1,900	\$ 1,900
Equipment	\$ 5,235	\$ 5,235	\$ 5,235	\$ 5,235	\$ 5,235	\$ 5,235
Truck	\$ 35,389	\$ 35,389	\$ 35,389	\$ 35,389	\$ 35,389	\$ 35,389
BOD2 Machine	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Less Accumulated Depreciation	\$ -	\$ 6,068	\$ 12,136	\$ 18,204	\$ 24,272	\$ 30,340
Total fixed	\$ 42,524	\$ 48,592	\$ 54,660	\$ 24,320	\$ 18,252	\$ 12,184
Total assets	\$ 375,538	\$ 323,735	\$ 311,551	\$ 308,650	\$ 382,709	\$ 517,514
Current Liabilities						
Accounts Payable		\$ -	\$ -	\$ -	\$ -	\$ -
Owner's Equity						
Initial Investment	\$ 375,538	\$ -	\$ -	\$ -	\$ -	\$ -
Retained earnings	\$ -	\$ 323,735	\$ 311,551	\$ 308,650	\$ 382,709	\$ 517,514
Total Owner's Equity	\$ 375,538	\$ 323,735	\$ 311,551	\$ 308,650	\$ 382,709	\$ 517,514
Total Liability and OE	\$ 375,538	\$ 323,735	\$ 311,551	\$ 308,650	\$ 382,709	\$ 517,514

Analysis	Year 1	Year 2	Year 3	Year 4	Year 5	Average
Fixed assets turnover ratio	6.62	6.66	6.81	17.59	26.96	12.93

Table 16. Balance Statement per year

Appendix I. Detailed Social Media Marketing Plan by Platform

Platform	Description
LinkedIn	LinkedIn will serve as EduMOD's primary platform for professional engagement and brand positioning. Content will highlight product benefits, purchasing opportunities, upcoming seminars, and STEM event participation. Thought leadership articles on STEM education trends, district pilot success stories, and partner school recognition will further enhance credibility. Paid ad campaigns will specifically target STEM Coordinators, District Superintendents, and School Principals. Posting frequency will be maintained at 2–3 times per week, featuring a mix of photos, articles, infographics, and short video clips to drive professional interaction.
X (formerly Twitter)	X will function as EduMOD's fast-paced engagement channel, ideal for real-time updates and event coverage. The platform will be used to live tweet from STEM events and conferences, share quick stats and infographics about cosmic ray science, and amplify educator mentions. Active participation in trending hashtags such as #STEMeducation, #K12EdTech, and #NGSS will strengthen visibility among education professionals. Posting frequency: 1–2 tweets per day.
Youtube	YouTube, the most widely used social media platform across all U.S. age groups (Pew Research Center, 2024), will serve as EduMOD's primary hub for visual storytelling. Content will include product demonstration videos, classroom walkthroughs, teacher testimonials, and explainer videos about muon detection science. Regular video publishing will boost EduMOD's SEO, enhance brand authority, and support learning engagement from both educators and students. Posting frequency: 2–3 videos per month.
Instagram	Instagram will strengthen EduMOD's community storytelling and authentic brand image. Focus areas will include behind-the-scenes content from STEM Family Nights, student project showcases, and reels summarizing participation in STEM events. Priority will be given to authentic classroom photography, real teacher and student moments, and dynamic Stories updates. Posts and reels will be shared 3–4 times per week to foster strong emotional connections with students, parents, and educator.

Facebook	Facebook will target PTAs, parent groups, and broader local school communities. Content will mirror Instagram’s visual storytelling but extend to longer, informative posts explaining EduMOD’s grant support services, pilot district success stories, and event invitations. As Facebook remains one of the most widely used platforms in the U.S. (alongside YouTube), paid Facebook ads will be optimized to reach local school district communities and education-focused groups. Posting frequency: 3 times per week.
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Table 17. Details Social Media Plan

Appendix J. Timing and Recipients of Personalised Package Delivery

To optimise communication effectiveness and increase stakeholder engagement, the personalised letter and parcel strategy will be implemented in two phases as follows:

Phase 1. Pre-Presentation (School-level target):	
Recipients:	Principals Heads of Science Departments STEM Teachers
Objectives	Create initial attention and interest Prepare recipients psychologically before the product presentation Increase EduMOD demo attendance
Implementation basis	School-level recipients are individuals who directly use EduMOD and are likely to attend presentations. Early exposure through physical parcels helps them become more familiar with the product, thereby approaching the demo with a proactive and positive mindset.
Phase 2. Post-Presentation Mailing (District Level):	
Audience	Superintendent Director of Curriculum Director of Technology
Goal	Reinforce information after the presentation Increase the likelihood of conversion to a pilot or purchase decision Save costs by sending only to those who have participated and expressed interest
Implementation Basis	Ultimate decision-making authority typically resides at the district level. Sending post-presentation mailings helps EduMOD avoid overreaching while focusing resources on those who have the capacity to make substantive decisions and have had prior exposure to the product.

Table 18. Package Delivery Details